Chapter 26 Molecules and Clusters

团簇

26.1 Atomic orbital

electronic wave functions in atoms

↔atomic orbitals (AOs)

1s and 2s orbitals

$$\psi_{100} = R_{10}Y_{00} = \frac{1}{\sqrt{4\pi}}\left(\frac{Z}{a_0}\right)^{3/2} 2e^{-Zr/a_0}$$

$$\psi_{200} = R_{20}Y_{00} = \frac{1}{2\sqrt{8\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(2-\frac{Zr}{a_0}\right)e^{-Zr/2a_0}$$

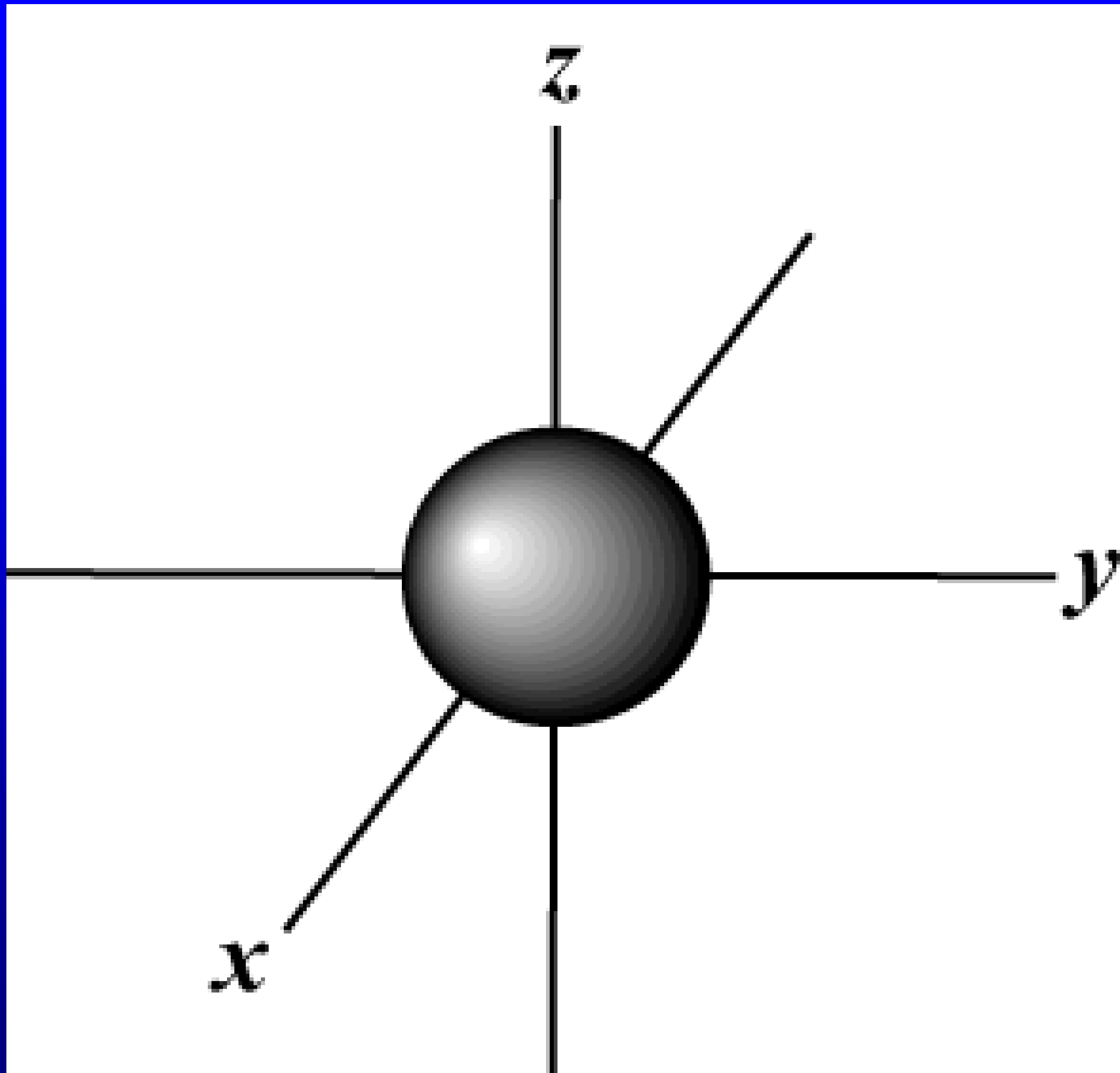
The probability distribution is (Z=1)

$$|\psi_{100}|^2 4\pi r^2 dr = 4\left(\frac{r}{a_0}\right)^2 e^{-2r/a_0} d\left(\frac{r}{a_0}\right)$$

$$\int_r^{\infty} 4\frac{r^2}{a_0^3} e^{-2r/a_0} dr = e^{-2r/a_0}\left(1+2\frac{r}{a_0}+2\frac{r^2}{a_0^2}\right)$$

$$=10\%, \leftrightarrow r = 2.6a_0$$

The probability of finding electron outside the surface of radius r


z
y
x

p orbitals

For n = 2, the radial part of wave function is

$$R_{21}(r) = \left(\frac{Z}{2a_0}\right)^{3/2} \frac{1}{\sqrt{3}} \frac{Zr}{a_0} e^{-Zr/2a_0}$$

The angular part is

common eigenfunctions of $\hat{L}^2$ and $\hat{L}_z$

$$Y_{1,0}(\theta,\varphi) = \sqrt{\frac{3}{4\pi}} \cos\theta$$

$$Y_{1,1}(\theta,\varphi) = -\sqrt{\frac{3}{8\pi}} \sin\theta \; e^{i\varphi}$$

$$Y_{1,-1}(\theta,\varphi) = \sqrt{\frac{3}{8\pi}} \sin\theta \; e^{-i\varphi}$$

The linear combination of angular part:

$$-\frac{1}{\sqrt{2}}\left(Y_{1,1}-Y_{1,-1}\right)=\left(\frac{3}{4\pi}\right)^{1/2}\sin\theta\cos\varphi=\left(\frac{3}{4\pi}\right)^{1/2}\frac{x}{r}$$

$$\frac{\mathrm{i}}{\sqrt{2}}\left(Y_{1,1}+Y_{1,-1}\right)=\left(\frac{3}{4\pi}\right)^{1/2}\sin\theta\sin\varphi=\left(\frac{3}{4\pi}\right)^{1/2}\frac{y}{r}$$

$$Y_{1,0}=\left(\frac{3}{4\pi}\right)^{1/2}\cos\theta=\left(\frac{3}{4\pi}\right)^{1/2}\frac{z}{r}$$

p_x, p_y, p_z orbitals

$$p_x = R_{21}(r)\left(\frac{3}{4\pi}\right)^{1/2}\frac{x}{r} \sim \quad xe^{-Zr/2a_0}$$

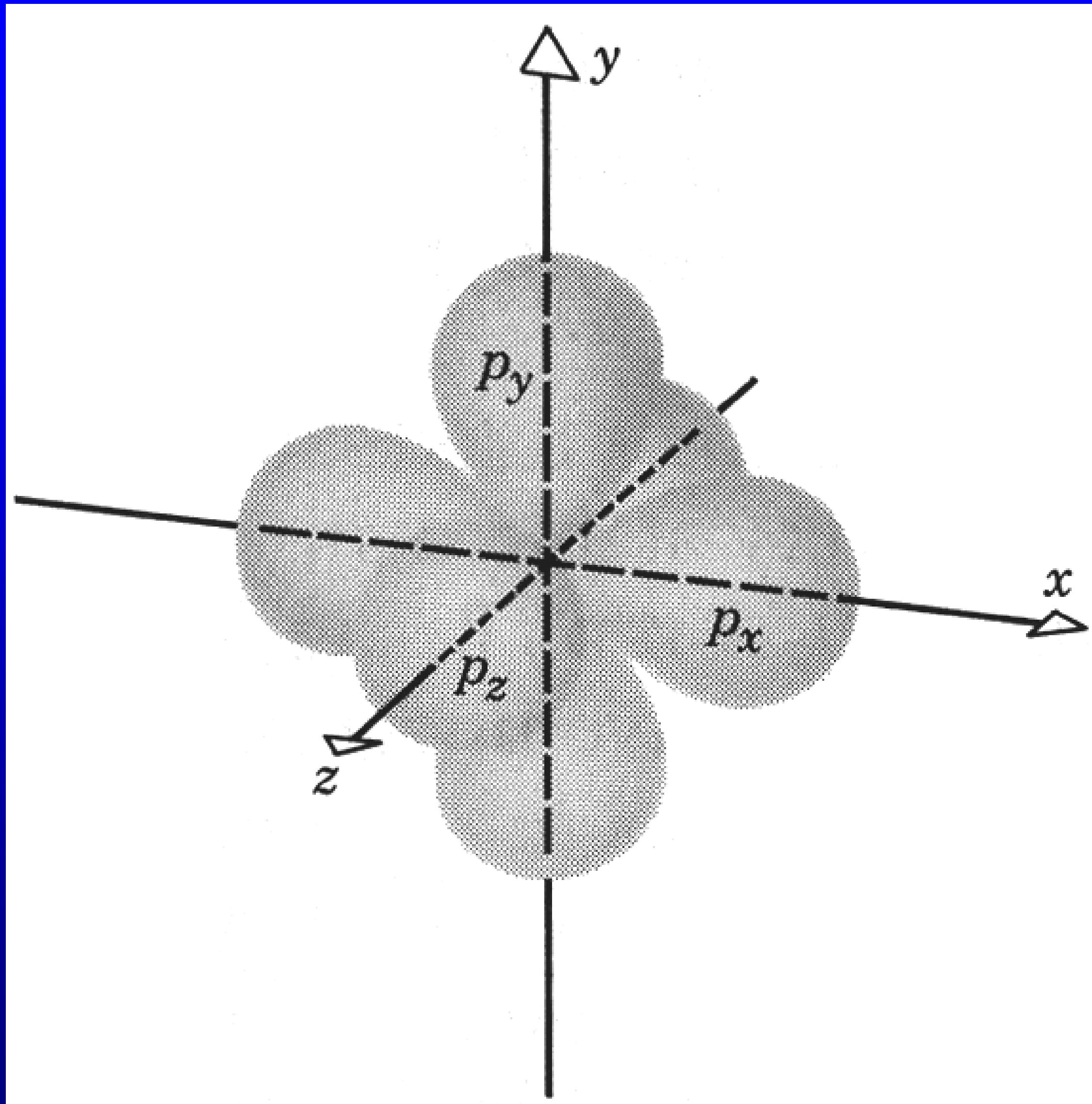
$$p_y = R_{21}(r)\left(\frac{3}{4\pi}\right)^{1/2}\frac{y}{r}$$

$$p_z = R_{21}(r)\left(\frac{3}{4\pi}\right)^{1/2}\frac{z}{r}$$

p_x common eigenfunction of $\hat{L}^2$ and $\hat{L}_x$

p_y common eigenfunction of $\hat{L}^2$ and $\hat{L}_y$

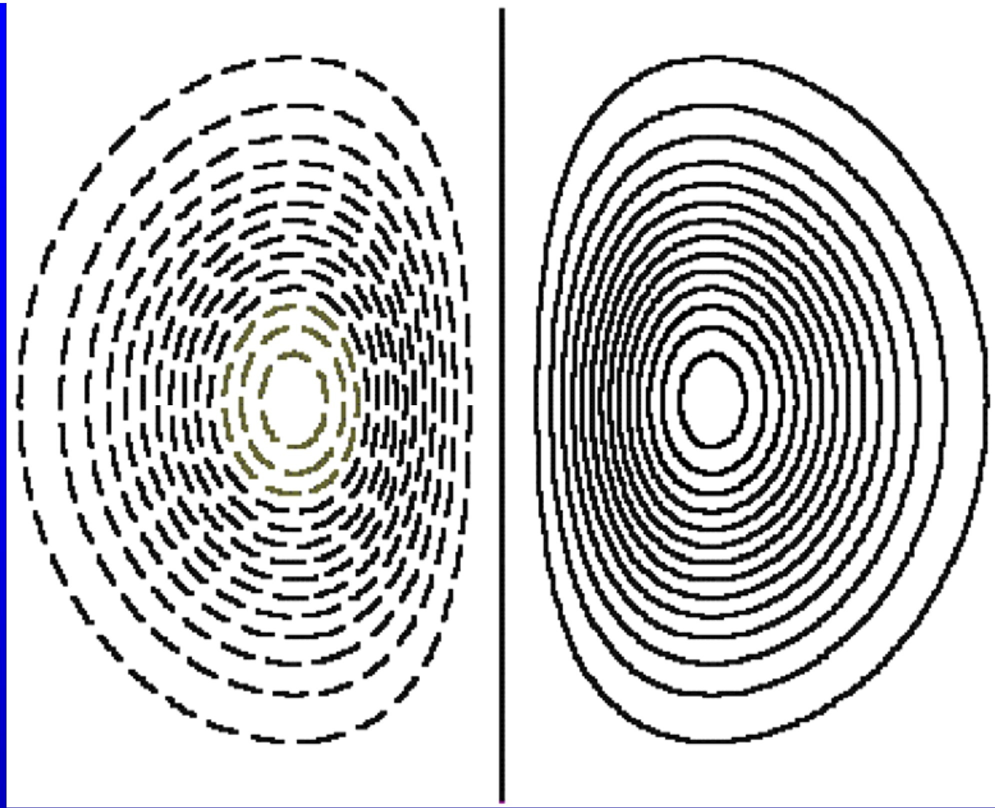
p_z common eigenfunction of $\hat{L}^2$ and $\hat{L}_z$



y
p_y
x
p_x
p_z
z

The probability density in x-z plane:

$$w_{2x}\Big|_{y=0} = \frac{1}{4\pi}\left(\frac{1}{2a_0}\right)^3\left(\frac{x}{a_0}\right)^2 e^{-\sqrt{x^2+z^2}/a_0}$$



$$w_{2x}\Big|_{y=0} = \frac{1}{4\pi}\left(\frac{1}{2a_0}\right)^3\left(\frac{x}{a_0}\right)^2 e^{-\sqrt{x^2+z^2}/a_0} \qquad p_x = R_{21}(r)\left(\frac{3}{4\pi}\right)^{1/2}\frac{x}{r}$$

Contours of w_{2x}

solid lines ---positive

dashed lines ----negative

sp hybridization

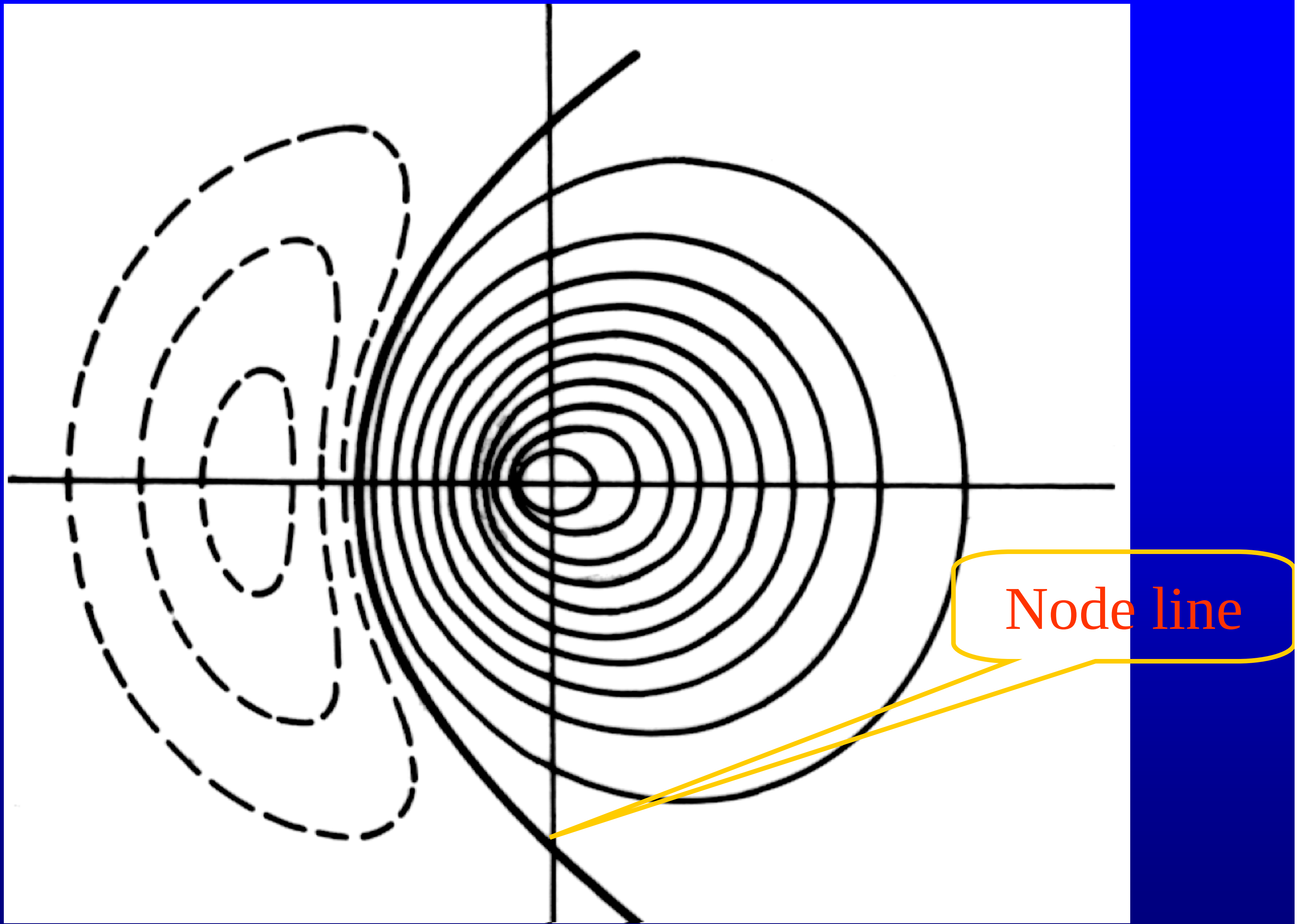
$$\frac{1}{\sqrt{2}}\left(\psi_{2s} \pm \psi_{2p}\right)$$

$$= \frac{1}{\sqrt{2}}\left[\left(\frac{1}{2a_0}\right)^{3/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}\frac{1}{\sqrt{4\pi}}\right.$$

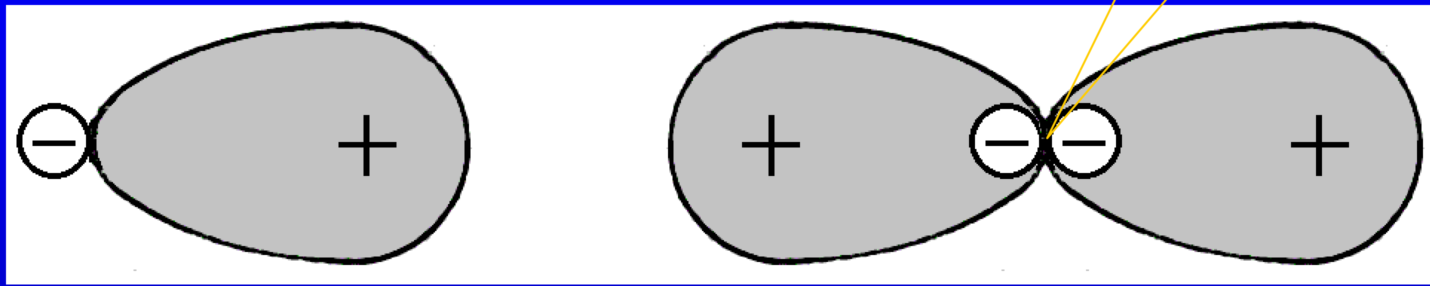
$$\left.\pm\left(\frac{1}{2a_0}\right)^{3/2}\frac{1}{\sqrt{3}}\left(\frac{r}{a_0}\right)e^{-r/2a_0}\left(\frac{3}{4\pi}\right)\frac{x}{r}\right]$$

$$= \frac{1}{\sqrt{2}}\left(\frac{1}{2a_0}\right)^{3/2}\frac{1}{\sqrt{4\pi}}\left(2-\frac{r}{a_0}\pm\frac{x}{a_0}\right)e^{-r/2a_0}$$

node line(+) in the *x*-*z* plane: $z^2 \mp 4a_0 x - 4a_0^2 = 0.$


Node line

Node point



Carbon atom

ground state configuration $1s^2 2s^2 2p^2$

CH_4 (methane) $1s^2 2s^2 2p^2 \longrightarrow 1s^2 2s^1 2p^3$

<u>hybridizations----linear combinations</u>

$$\psi_1 = \frac{1}{2}(s + p_x + p_y + p_z)$$

$$s = -\psi_{200}$$

$$\psi_2 = \frac{1}{2}(s + p_x - p_y - p_z)$$

sp^3 hybridization

$$\psi_3 = \frac{1}{2}(s - p_x + p_y - p_z)$$

The angular momentum is not well defined

$$\psi_4 = \frac{1}{2}(s - p_x - p_y + p_z)$$

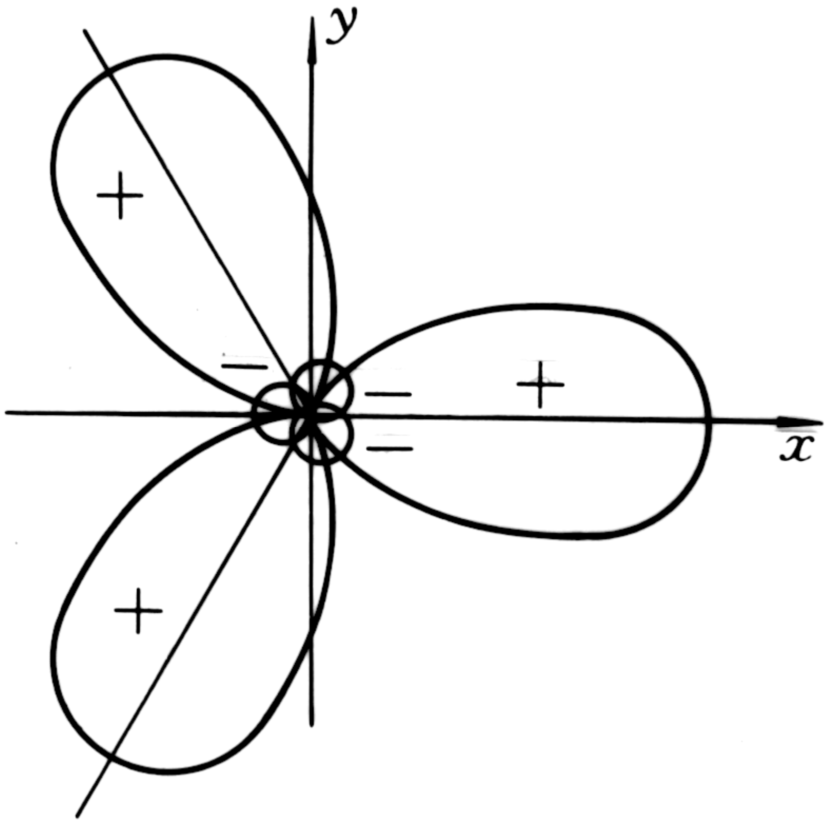
sp^2 hybridization

$$\psi_1 = \frac{1}{\sqrt{3}}(s + \sqrt{2}\,p_x)$$

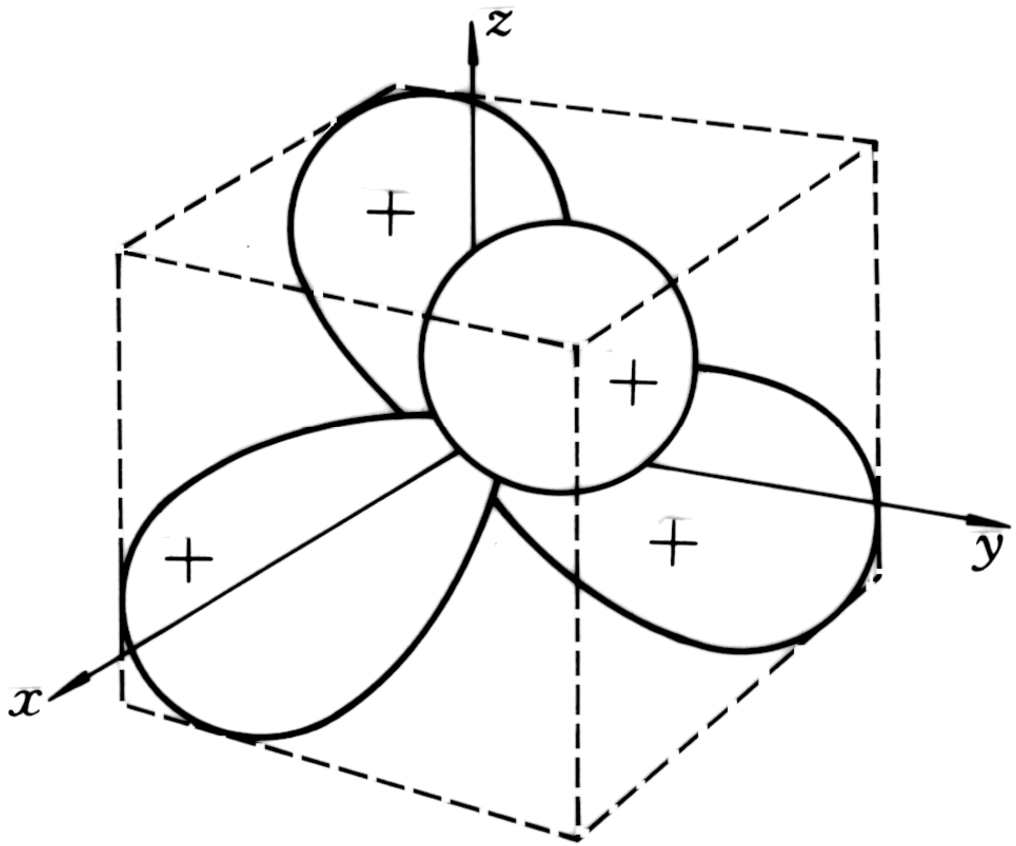
$$\psi_2 = \frac{1}{\sqrt{3}}(s - \frac{1}{\sqrt{2}}\,p_x - \sqrt{\frac{3}{2}}\,p_y)$$

$$\psi_3 = \frac{1}{\sqrt{3}}(s - \frac{1}{\sqrt{2}}\,p_x - \sqrt{\frac{3}{2}}\,p_y)$$

C_2H_4 (ethylene) $1s^22s^22p^2 \longrightarrow 1s^2\underline{2s^12p^3} \longrightarrow sp^2+p$



y
x
+
+
+
−
−
−

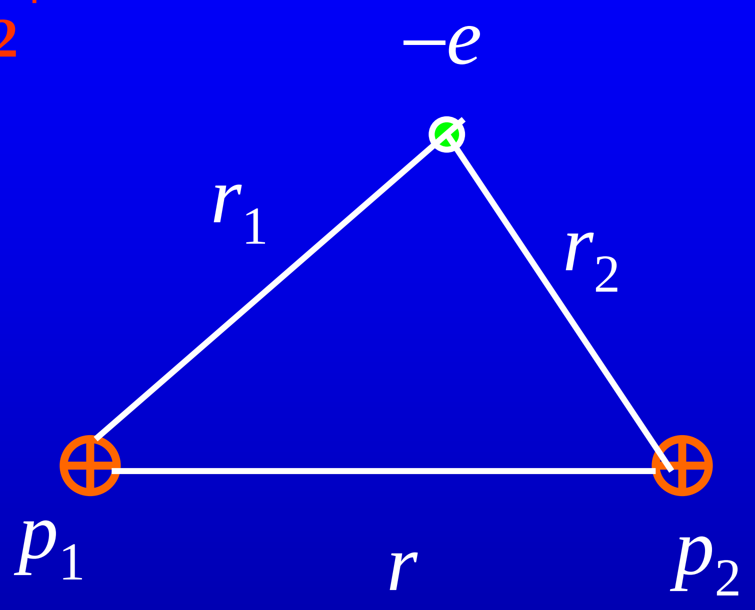


z
x
y
+
+
+
+

26.2 Molecular Orbital

Hydrogen molecule ion H_2^+

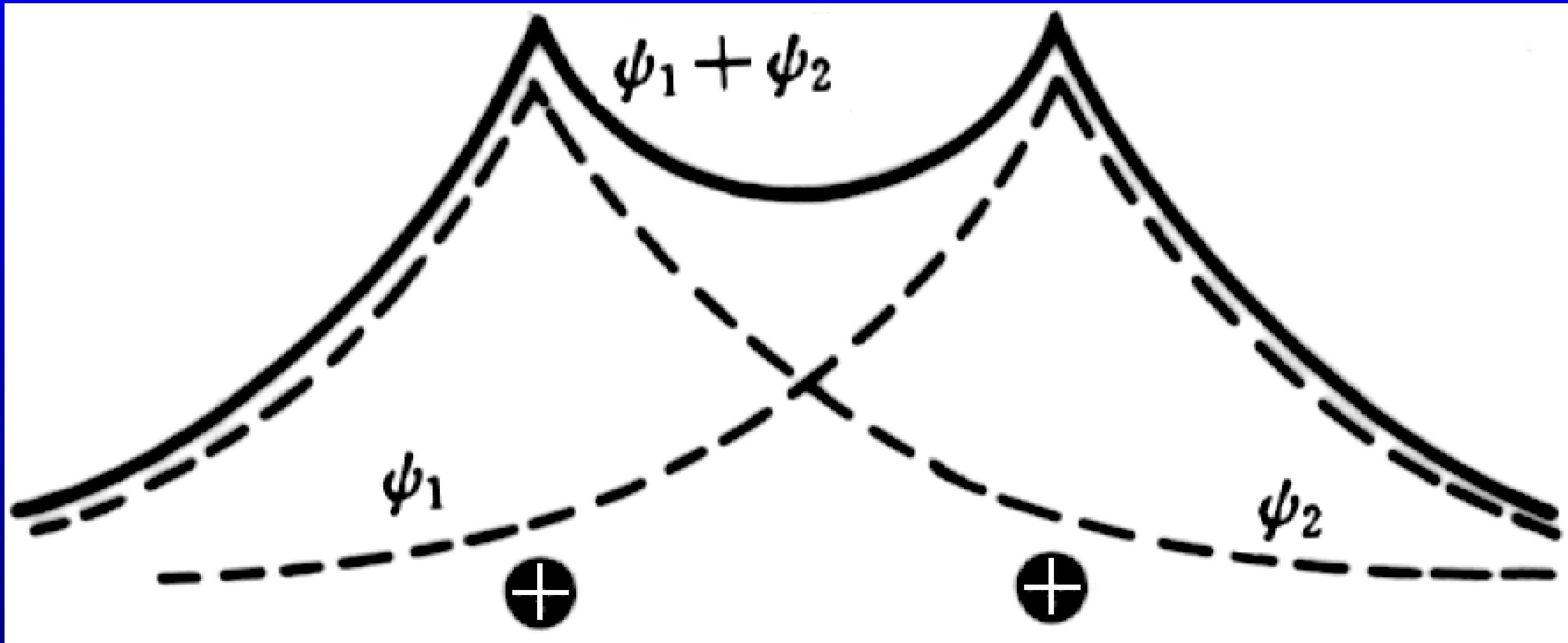
$$U = e_s^2\left(-\frac{1}{r_1}-\frac{1}{r_2}\right)+\frac{e_s^2}{r}.$$



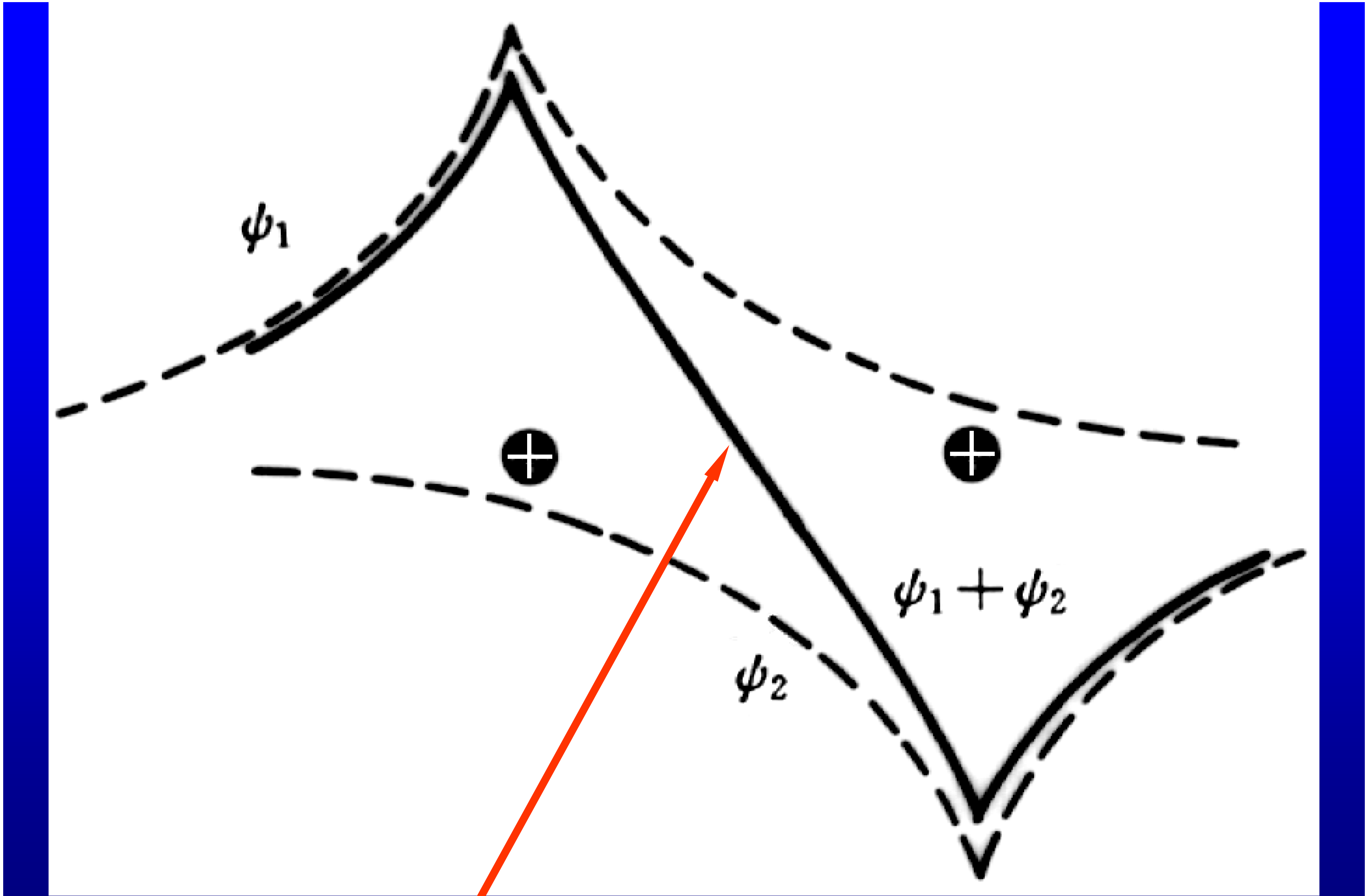
For two far away protons and an electron

$$\psi_{\text{even}} = \psi_1 + \psi_2$$

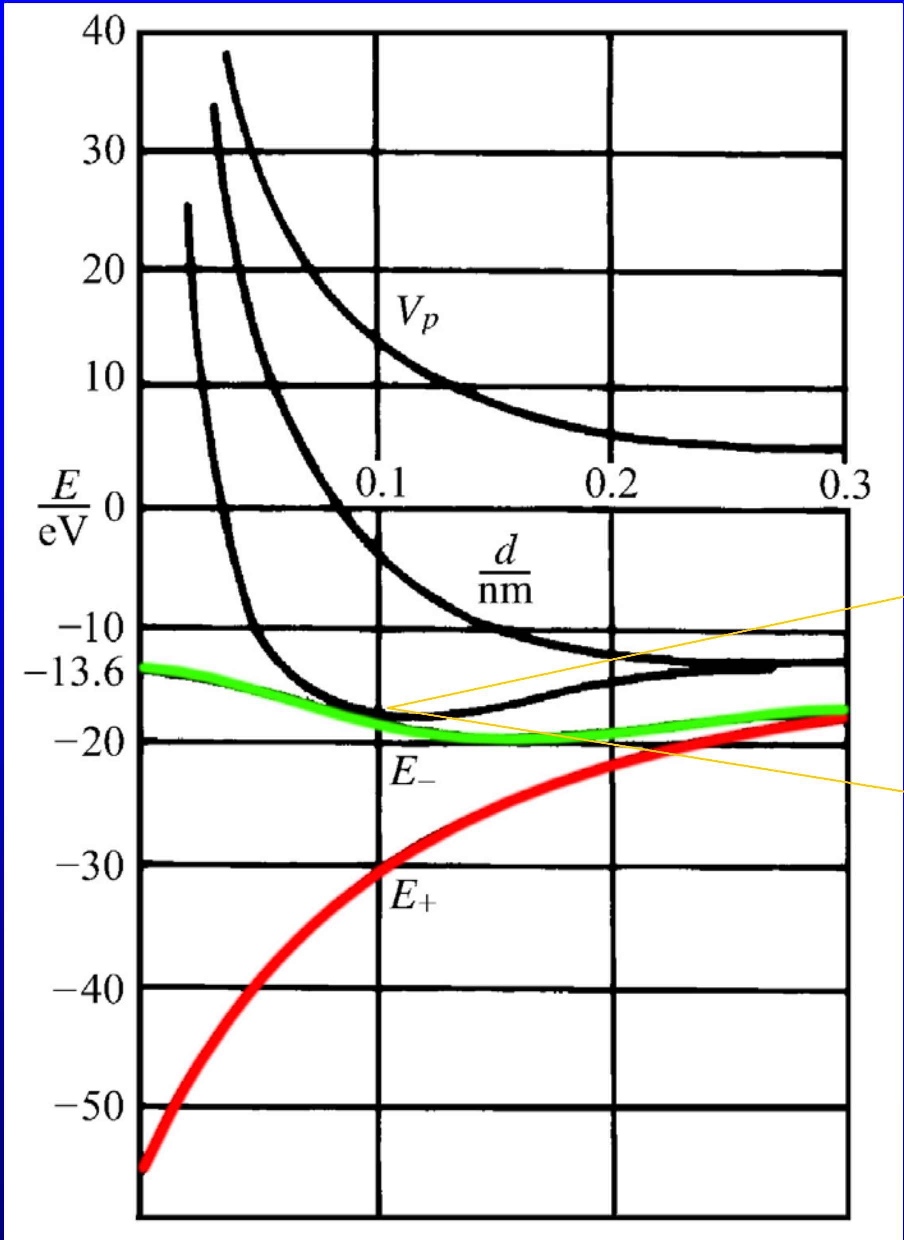
$$\psi_{\text{odd}} = \psi_1 - \psi_2$$



hydrogen-like atom with $Z = 2$



2p state



r_0= 0.106nm,
energy –16.3eV

–2.648eV relative to the H atom ground state.

$$r_0 \approx 2a_0$$

Energy consideration

$$E_{\pm}(\infty) = -13.6\,eV$$

$$E_{+}(0) = -13.6 \times Z^2 = -54.4\,eV$$

$$E_{-}(0) = -13.6 \times \frac{Z^2}{n^2} = -13.6\,eV$$

$$E_{\infty} - E_{\min} = 2.7\,\mathrm{eV}$$

binding energy

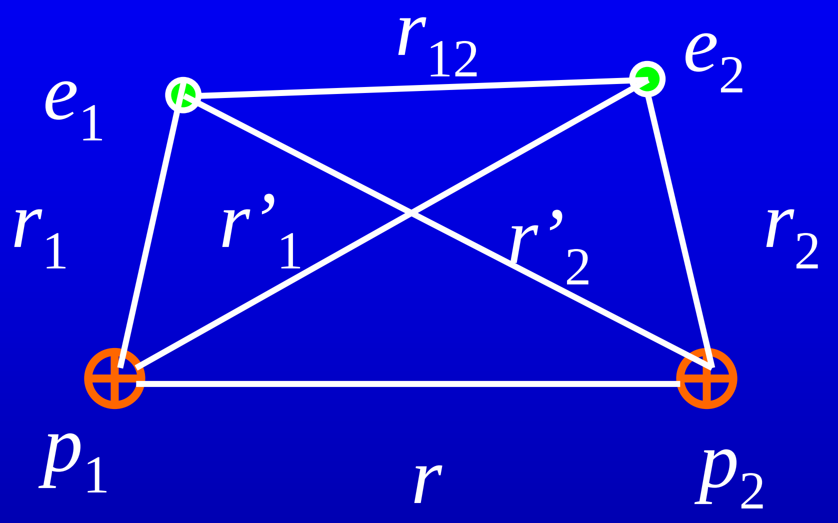
$$\psi_{\text{even}} = \psi_1 + \psi_2$$ bonding orbital

$$\psi_{\text{odd}} = \psi_1 - \psi_2$$ antibonding orbital

molecular orbitals (MOs)

LCAO linear combination of atomic orbitals

The H_2 molecule and the covalent bond



$$U = e_S^2\left(-\frac{1}{r_1}-\frac{1}{r_1'}-\frac{1}{r_2}-\frac{1}{r_2'}\right)+\frac{e_S^2}{r}.$$

$$H_2 = H_2^+ + 1\,\text{electron} \longrightarrow \sigma_g 1s$$

$\sigma_g 1s$ molecular orbital

H_2 $(\sigma_g 1s)^2$ configuration

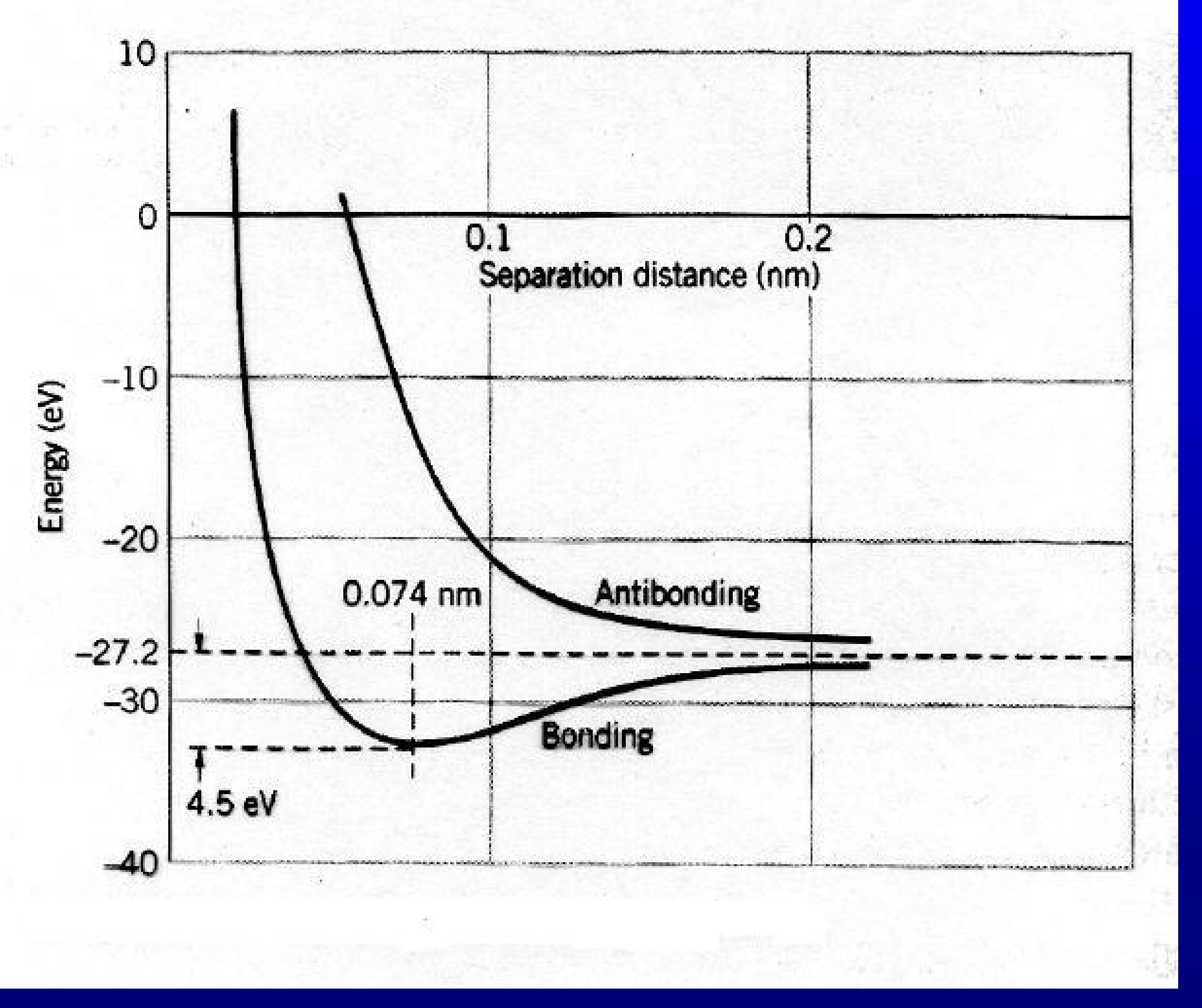
Both two electrons are in a bonding state, but with opposite spin.

even

antibonding

$(\sigma_g 1s)(\sigma_u^* 1s)$

odd



minimum energy –31.7eV

separation $1.40a_0$ = 0.074nm,
the binding energy 4.75eV,
bonding energy 423kJ/mol

exchange interaction   a quantum effect

Molecular orbitals of diatomic molecules

$$m = \quad 0, \pm 1, \pm 2, \pm 3, \cdots$$

$$|m| = \quad 0, \ 1, \quad 2, \quad 3, \cdots$$

symbols $\sigma, \ \pi, \ \delta, \ \phi, \cdots$

$\sigma n\ell, \ \ \pi n\ell$

German gerade -----even

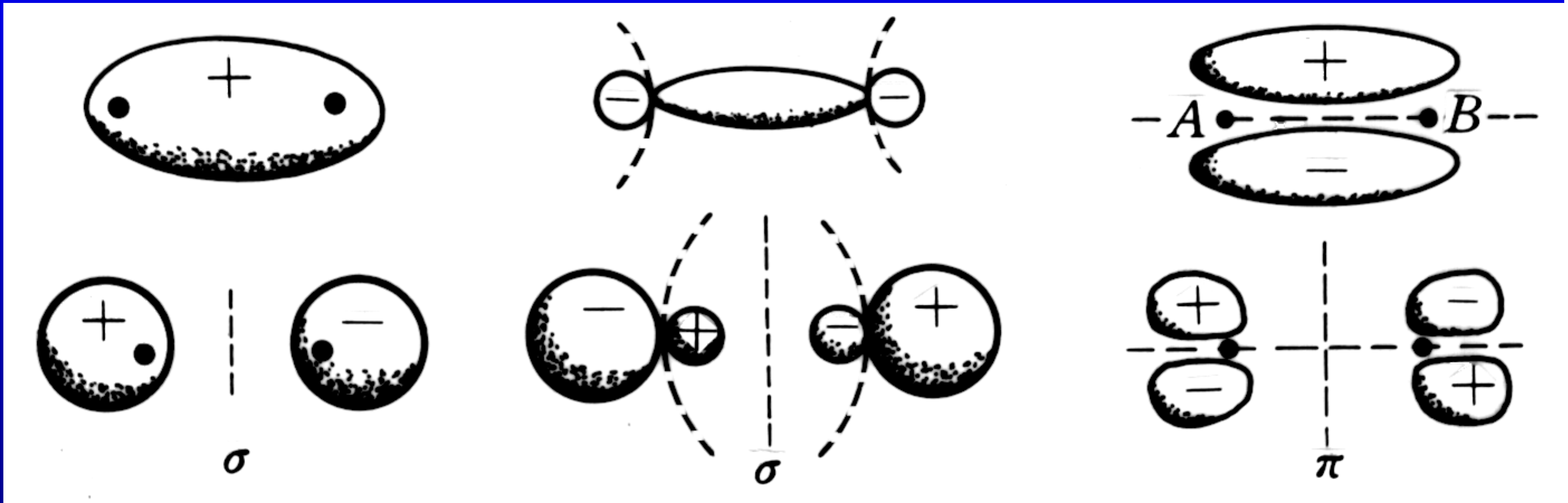
ungerade----odd

$\psi_{\text{even}} \rightarrow g$ state

$\psi_{\text{odd}} \rightarrow u$ state

$\sigma_g, \sigma_u, \pi_g, \pi_u$

p-p covalent bond



σ_g ns

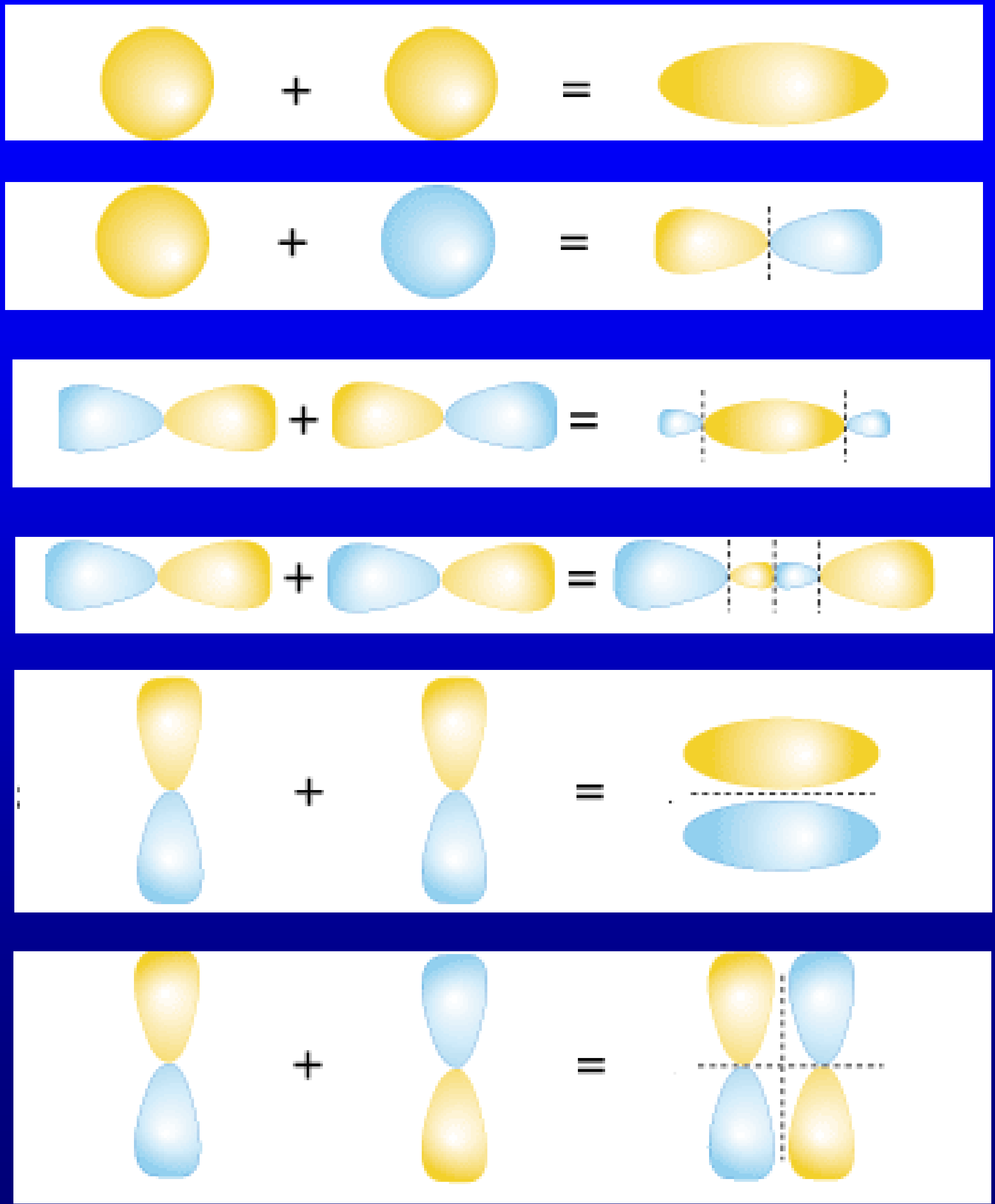
σ_u^* ns

σ_g np

σ_u^* np

π_u np

π_g^* np



σ_g ns

σ_u^* ns

σ_g np

σ_u^* np

π_u np

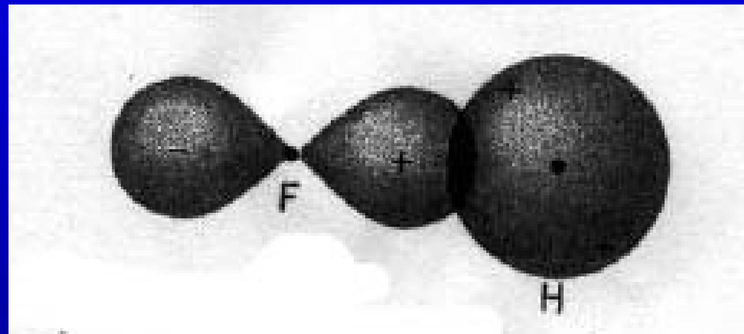
π_g^* np

s-p molecular bond

The configuration of fluorine (F) is $1s^2\,2s^2\,2p^5$

	s	p		
L	↑↓	↑↓	↑↓	↑
K	↑↓			

stable HF molecule



Hund's rule: The resultant spin of the ground state of atoms has the largest possible value compatible with the exclusion principle.

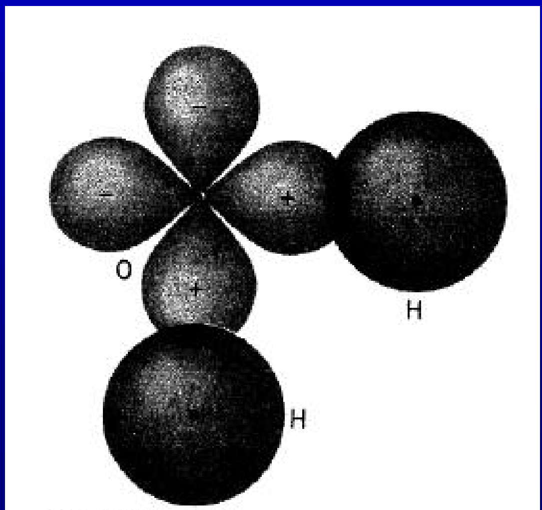
s-p directed bond

oxygen (O) has eight electrons and a configuration $1s^2\,2s^2\,2p^4$.

	s	p		
L	↑↓	↑↓	↑	↑
K	↑↓			

p_x p_y $p_z \longleftarrow\!\!\longrightarrow p_z$

pair



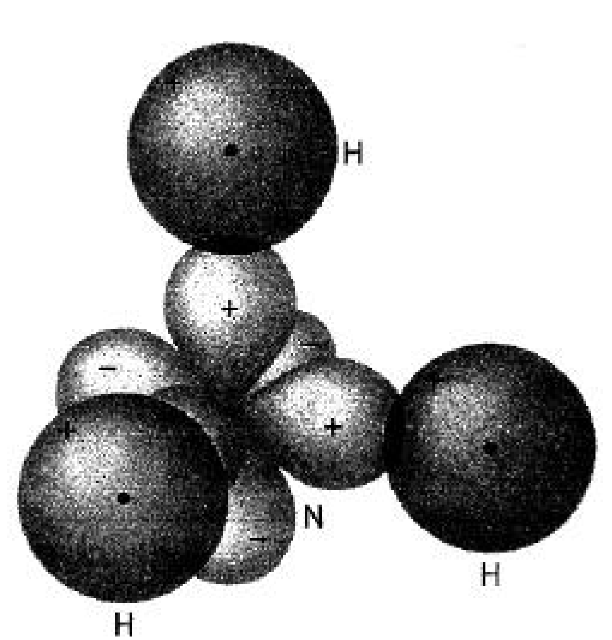
Coulomb repulsion of two H atoms 104.5°

H_2S (Hydrogen Sulfide) 92.3° 硫化氢

H_2Se (Hydrogen Selenide) 92.3° 硒化氢

Ammonia NH_3

Nitrogen (N) has seven electrons and configuration $1s^2 2s^2$ $2p^3$.

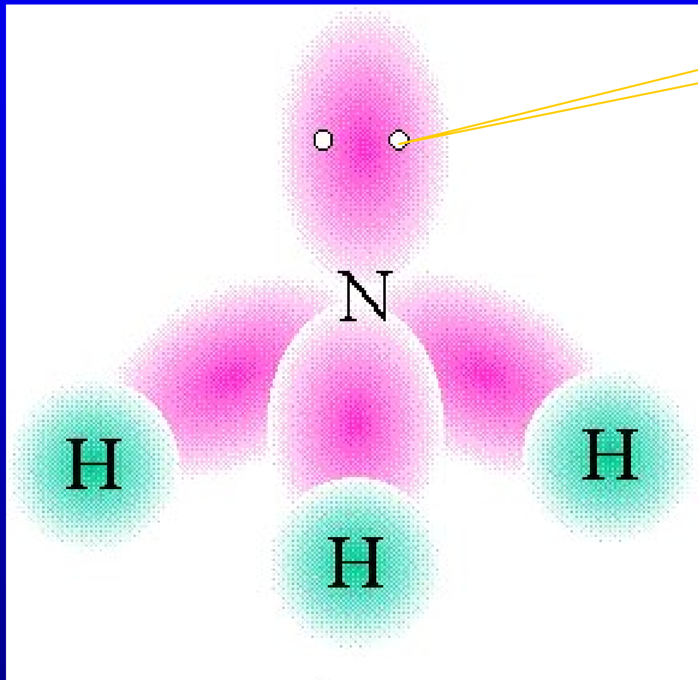


The bond angle is 107.3º

N $1s^2 2s^2 2p^3 \longrightarrow 1s^2 2s^1 (2p_x)^2 p_y p_z$;

sp^3 hybridization 109.5°

lone pair

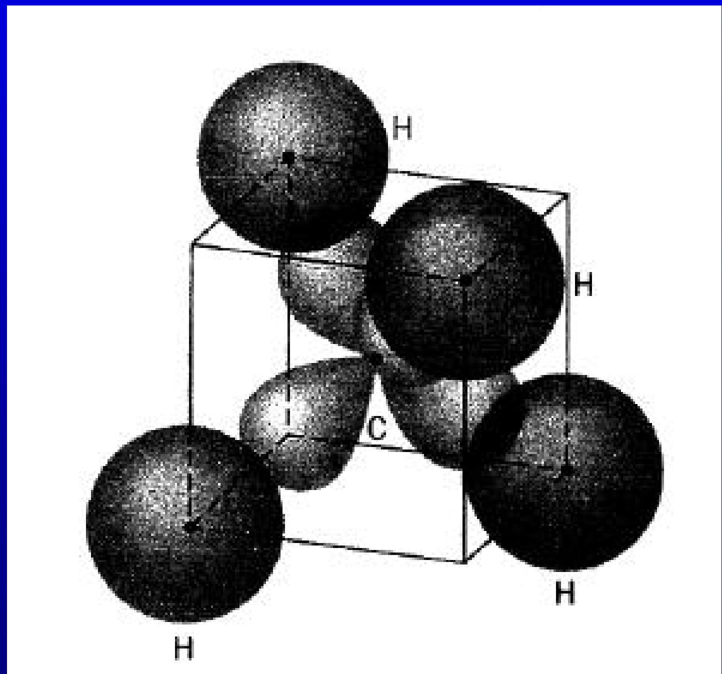


sp³ hybrid orbitals (tetrahedral)

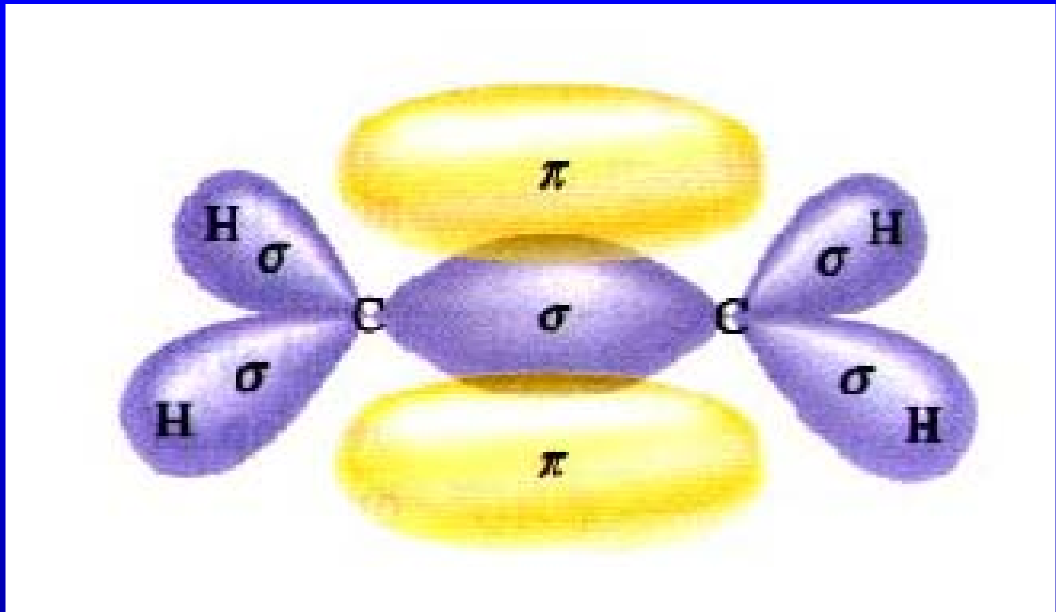
carbon and s–p hybrid orbital

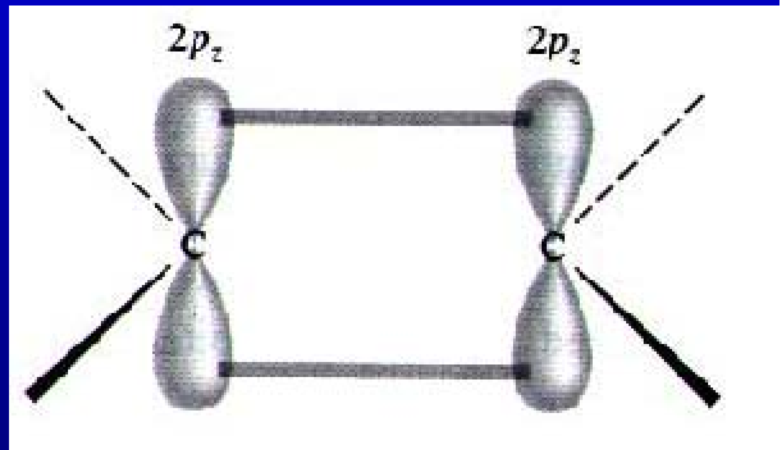
CH_4 (methane)　　C　$1s^2 2s^2 2p^2 \longrightarrow 1s^2 2s^1 2p^3$;

甲烷　　　　　　sp^3 hybridization



C_2H_4 (ethylene) $1s^2 2s^2 2p^2 \longrightarrow 1s^2 2s^1 2p^3 \longrightarrow sp^2 + p$



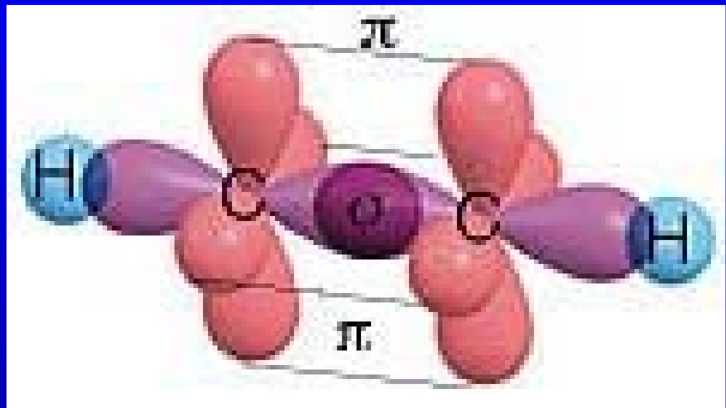


double σπ-bond

$H_2C = CH_2$

C_2H_2 (acetylene) $1s^2 2s^2 2p^2 \longrightarrow 1s^2 2s^1 2p^3 \longrightarrow sp + p^2$

乙炔



triple bond $1\sigma + 2\pi$

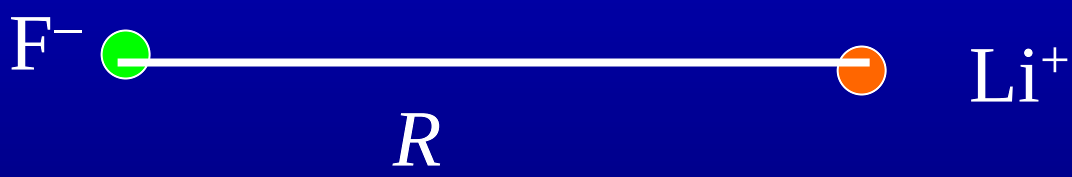
26.3 Ionic bonding

LiF lithium fluoride

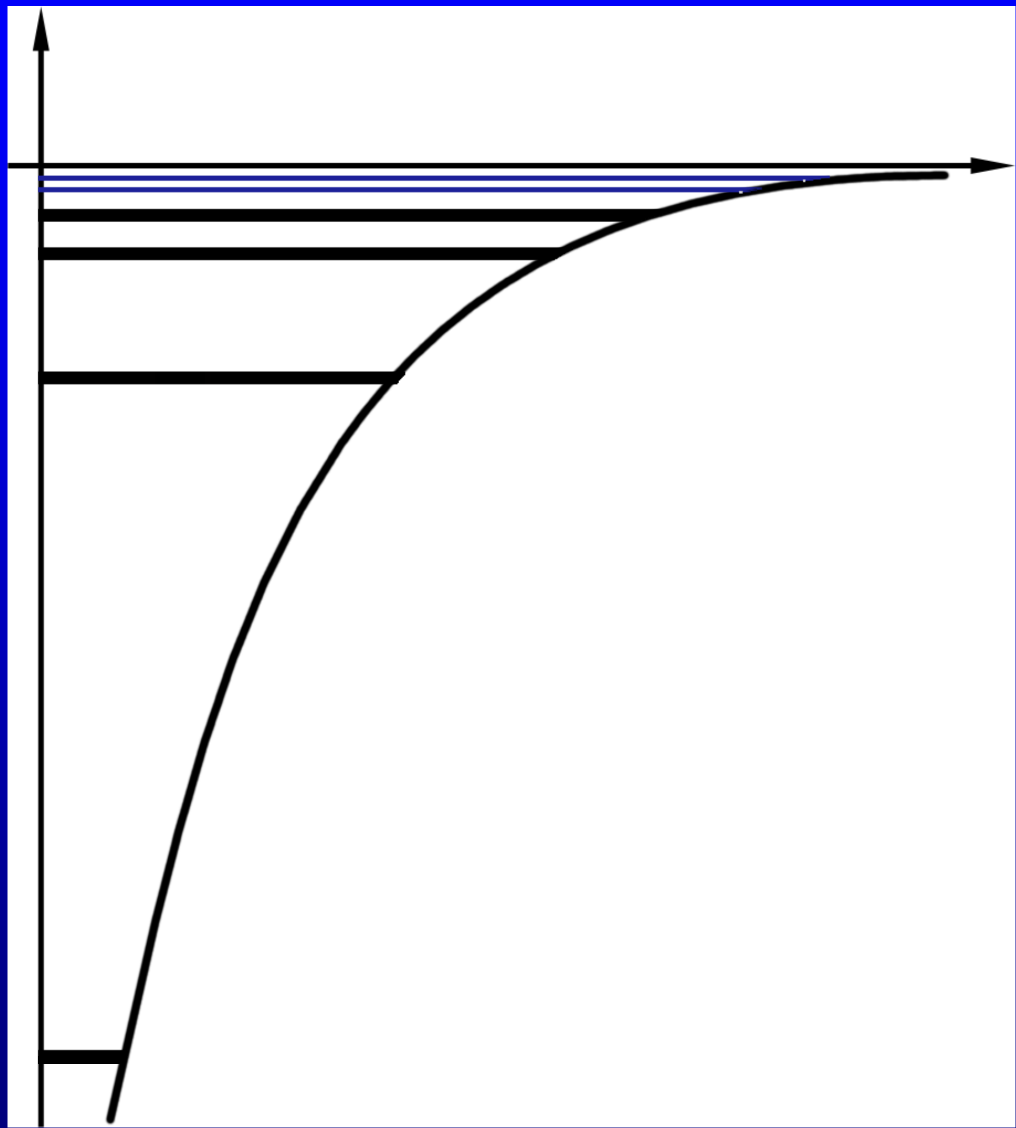
Ionization energy of lithium is 5.4 eV

Electron affinity of the fluorine is 3.5eV.

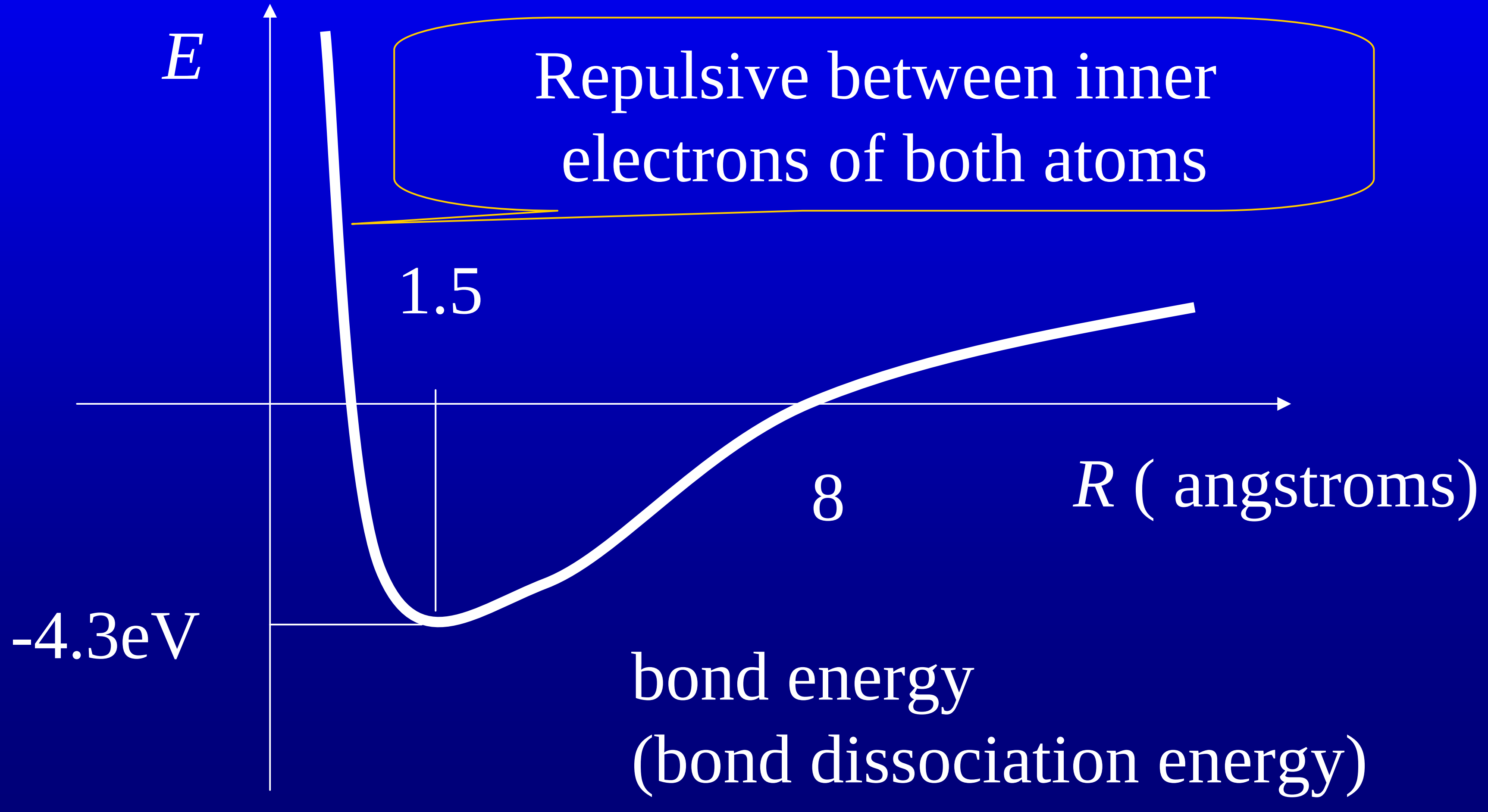
The net energy needed to remove the electron from lithium and attach it to the fluorine is 1.8 eV.



$$R = 8\,\mathring{A}, \quad \frac{1}{4\pi\varepsilon_0}\frac{e^2}{R} = 1.8\,\text{eV}.$$



Hence, at distance less than 8 angstrom, the out electron of the lithium can reach a lower energy state by ‘jumping’ over to the fluorine atom.



Ionic or covalent?

Generally
homonuclear diatomic molecules are formed through covalent bonds.
While in heteronuclear molecules there is no "pure" covalent bond.
A finite electric dipole is indication of ionic bond.

1. The measurement of the electric dipole moment.

In NaCl the separation is 2.36Å

$$p = 1.60\times10^{-19}\,\mathrm{C}\times 2.36\times10^{-10}\,m$$
$$= 3.78\times10^{-29}\,\mathrm{C\cdot m}$$

But the measured electric dipole moment is 3.00 $\times$ 10 $^{-29}$C ·m

percent ionicity is 3.00/3.78 = 79.4%

mostly ionic bond

In HCl ,

$$p = 1.60\times10^{-19}\ \mathrm{C}\times1.28\times10^{-10}\,m$$

$$= 2.05\times10^{-29}\ \mathrm{C\cdot m}$$

the measured electric dipole moment is $0.36\times10^{-29}\,\mathrm{C\cdot m}$ $p/p_{\mathrm{meas}}=18\%$. mostly covalent.

2. The electronegativity

----------the quantity that measures the ability of an atom in a molecule to attract or donate an electron.

Mulliken electronegativity

$$\chi = \frac{1}{3.17} \times \frac{1}{2}(\mathrm{IE} + \mathrm{EA})$$

(in unit eV)

IE ionization energy
EA electron affinity

For Li, $\chi = 1.$

The degree of ionicity of f_i of a molecular bond between atoms A and B depends on the difference between their electronegativity.

$$f_i = 1 - \exp\left[-\frac{(\chi_A - \chi_B)^2}{4}\right]$$

difference between electronegativity

zero pure covalent

small polar covalent 极性共价键

large ionic about >2

Intermolecular force:

Van der Waals force

dipole-dipole interaction ($\sim r^{-3}$)

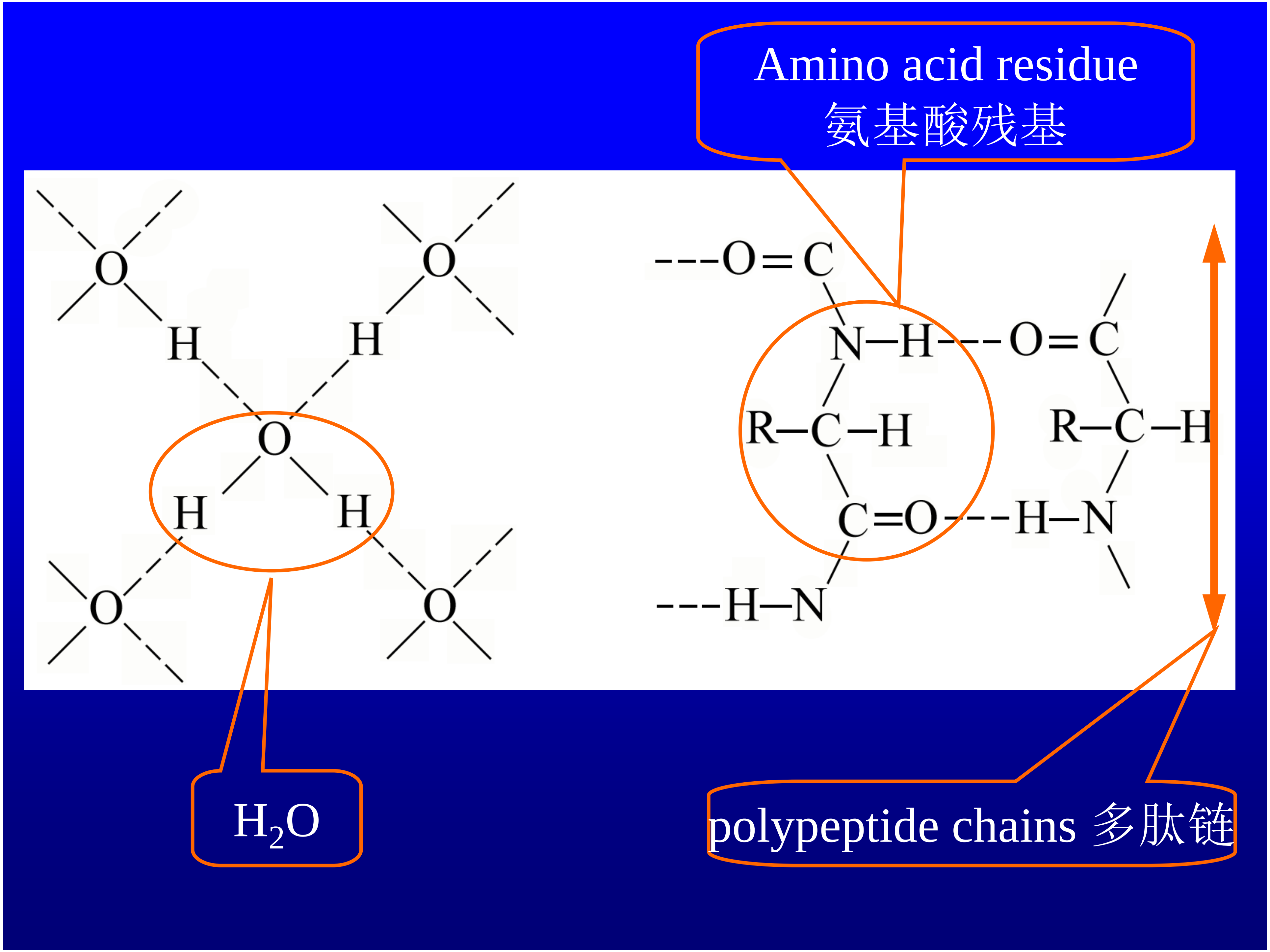
induced dipole force ($\sim r^{-7}$)

<< hydrogen bonds ($5 \sim 50 kJ \cdot mol^{-1}$)

$10^{-1} \sim 10^{-2} eV$

F O N strong electronegativity

<< chemical bonds ($200 \sim 500 kJ \cdot mol^{-1}$ several eV)


Amino acid residue
氨基酸残基
H2O
polypeptide chains 多肽链
---O=C
N–H---O=C
R–C–H
R–C–H
C=O---H–N
---H–N

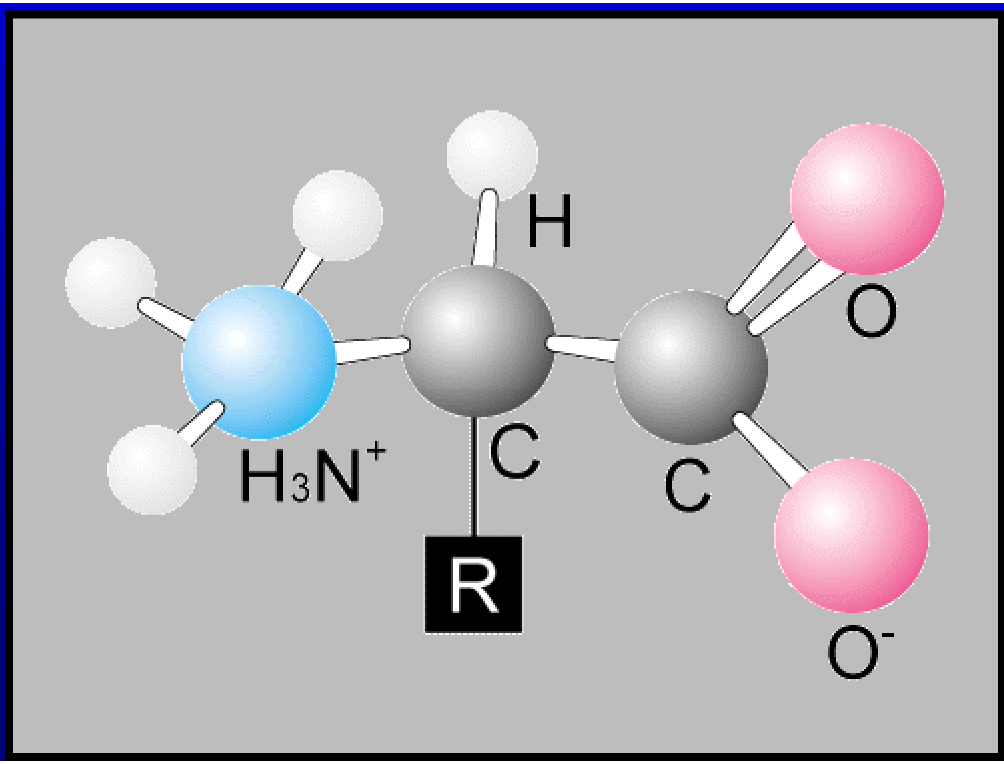
amino acid

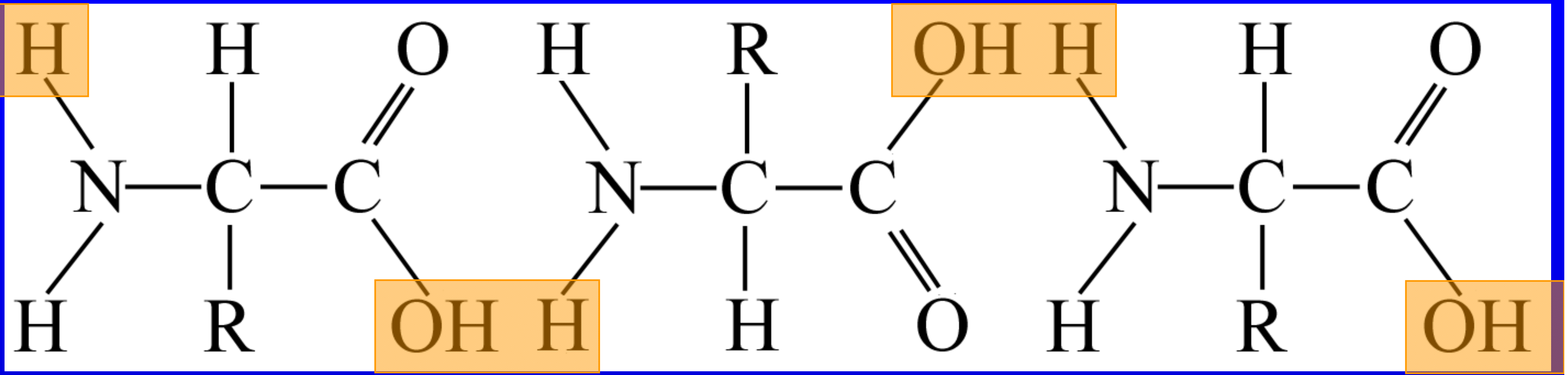
(side chain, R group)侧链,R基

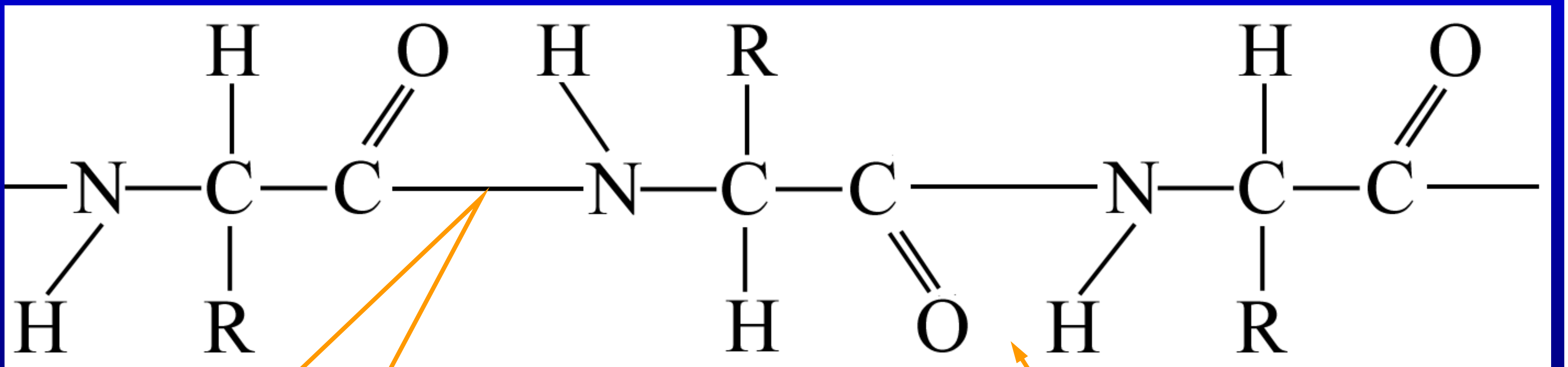
$$
\begin{array}{c}
\text{R} \\
| \\
\text{(Amino group)}H_2N\text{—}\mathrm{C}\text{—}\mathrm{COOH}\text{(Carboxyl group)} \\
| \\
\text{H}
\end{array}
$$

氨基

羧基



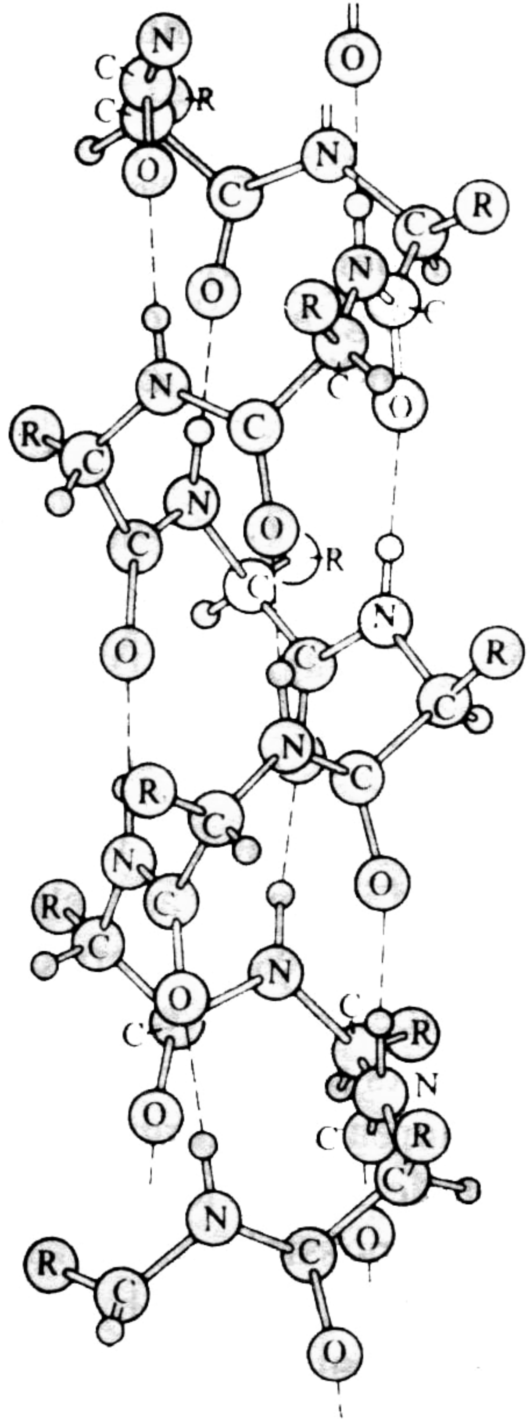




peptide bond
肽键

polypeptide chain
多肽链

Hydrogen bond
~ 3.6 amino acids
a complete turn



N
C
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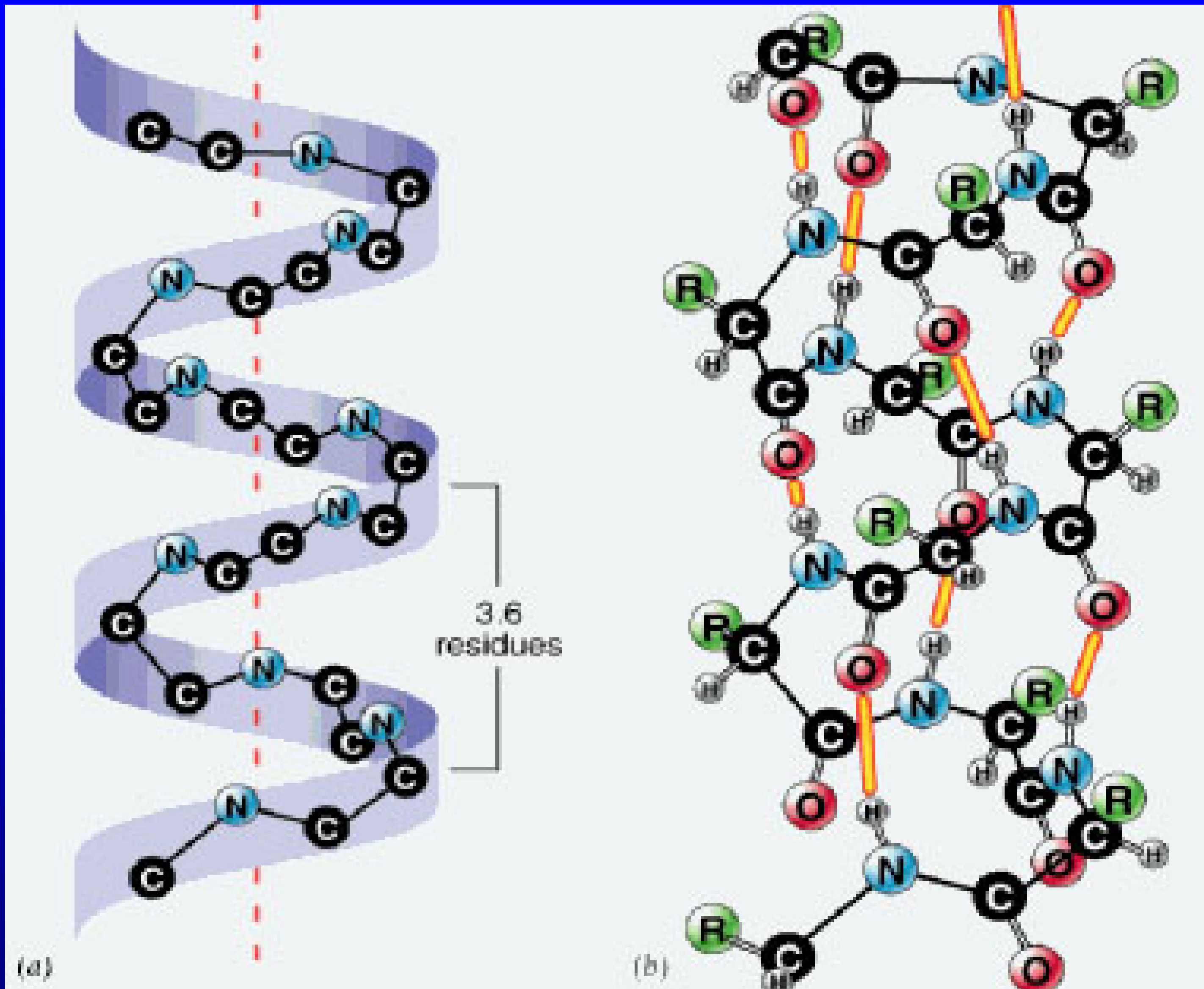
O

α-helix


3.6
residues
(a)
(b)

The primary structure ~ sequence of amino acids

The secondary structure ~ spatial organization of the protein chain or some part of it

The tertiary structure ~ Folding of protein

The enzyme is globular protein.

Assignment: 26.1,26.2,

Pauling scale

the most electronegative element (fluorine 氟) is given an electronegativity value of 4.0

(The least electronegative element (francium钫) has a value of 0.7.)

bond energy

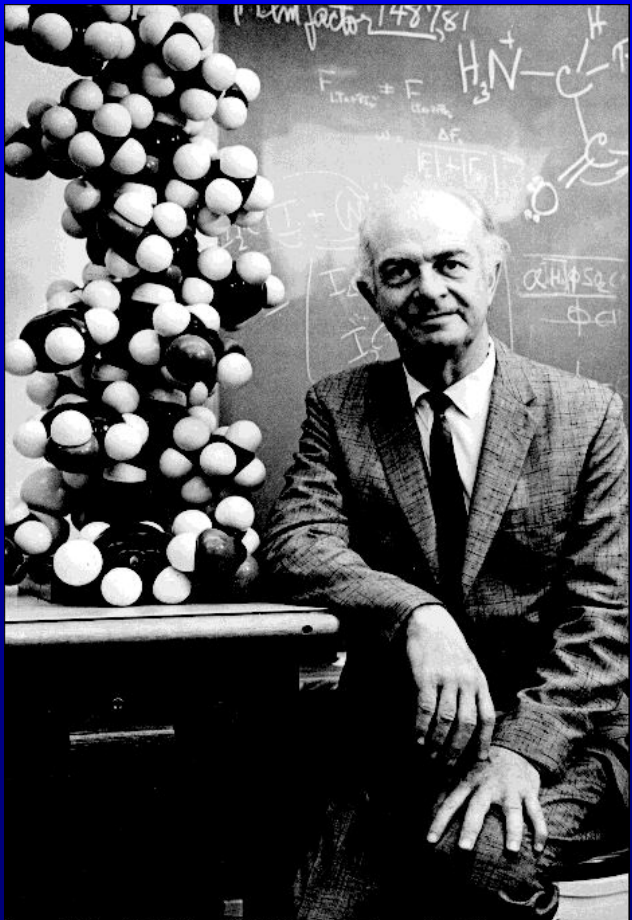
$$E(\mathrm{AB}) = \sqrt{E(\mathrm{AA}) \cdot E(\mathrm{BB})} + 96.48(\chi_{\mathrm{A}} - \chi_{\mathrm{B}})^2$$

E(AB)   bond energy in molecule AB

E(AA) (E(BB))   bond energy in homonuclear molecule AA (BB)   in unit $\mathrm{kJ \cdot mol^{-1}}$

$$\chi = 3.48 \cdot \left[\frac{(\mathrm{IE} + \mathrm{EA})}{2} - 0.602 \right]$$

(IE and EA in unit $\mathrm{MJ \cdot mol^{-1}}$)



Linus Carl Pauling

The Nobel Prize in Chemistry 1954

The Nobel Peace Prize 1962

Denaturation of Proteins by heat

Heat can be used to disrupt hydrogen bonds and non-polar hydrophobic interactions. This occurs because heat increases the kinetic energy and causes the molecules to vibrate so rapidly and violently that the bonds are disrupted. The proteins in eggs denature and coagulate during cooking. Other foods are cooked to denature the proteins to make it easier for enzymes to digest them. Medical supplies and instruments are sterilized by heating to denature proteins in bacteria and thus destroy the bacteria.

26.4 Molecular vibrations and rotations

The potential energy in a diatomic molecule:

$$U(r) = U(r_0) + U'(r_0)(r - r_0) + \frac{1}{2}U''(r_0)(r - r_0)^2 + \cdots$$

$$\approx U_0 + \frac{1}{2}k(r - r_0)^2$$

The vibration energy levels

$$E_v = \hbar\omega\left(n + \frac{1}{2}\right), \ \omega^2 = \frac{k}{\mu}$$

μ---reduced mass

Selection rule:

$$\Delta n = \pm 1.$$

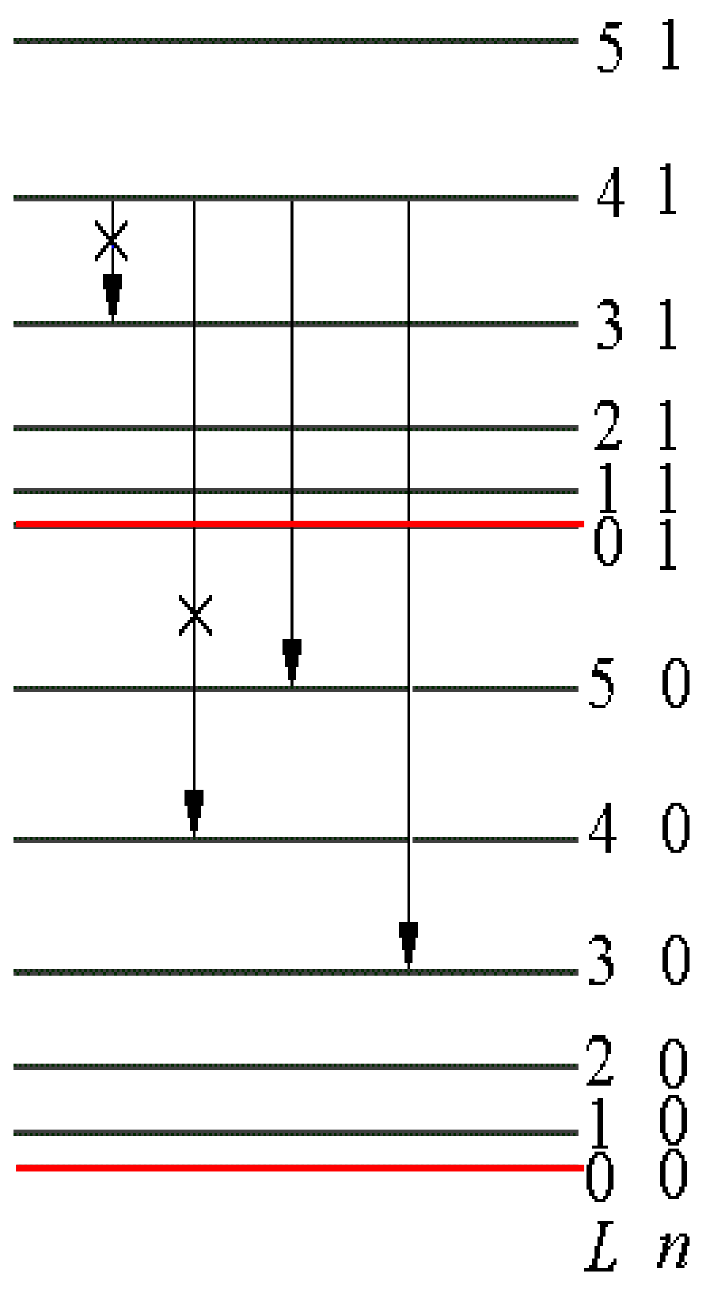
$$\hat{H}_r = \frac{\hat{L}^2}{2I}, \qquad I = \mu R^2$$

The rotation energy levels

$$L^2 = L(L+1)\hbar^2 \Rightarrow \quad E_r = \frac{L^2}{2I} = \frac{L(L+1)\hbar^2}{2\mu R^2}$$

Selection rule:

$$\Delta L = \pm 1.$$



$$\Delta E = E_{N,L} - E_{N-1,L\pm 1}$$

$$= \begin{cases} h\nu + \dfrac{\hbar^2}{mR^2}(-L-1) & L \to L+1 \\ h\nu + \dfrac{\hbar^2}{mR^2} L & L \to L-1 \end{cases}$$

Molecular spectra

"Characteristic fingerprint"

1.The absorption spectra of the atmosphere can be used to identify <u>trace amounts</u> of various pollutants and then to measure the purity of the air.

2. The absorption spectra of the interstellar dust can be used to identify the molecules that are present in interstellar space.

Assignment: 26.9, 26.11

26.5 Biological macromolecule

The main biological macromolecules are proteins, nucleic acids, polysaccharides and lipids(蛋白质、核酸、多糖和脂质).

Proteins and nucleic acids are sometimes referred to as informational macromolecules --they have precisely defined structures, and reflect the genetic flow of information (DNA —RNA —protein).

DNA (deoxyribonucleic acid)

Nitrogen base
含氮碱基

A G G C T T A

— S — P — S — P — S — P — S — P — S — P — S — P — S —

sugar(deoxyribose)
脱氧核糖

Phosphate
磷酸

A(adenine) 腺嘌呤， G(guanine) 鸟嘌呤
T(thymine) 胸腺嘧啶 C(cytosine) 胞嘧啶

adeno: latin for “gland” guano:dung of sea birds or bats
thymus：胸腺 cyt, cyto pertaining to cells

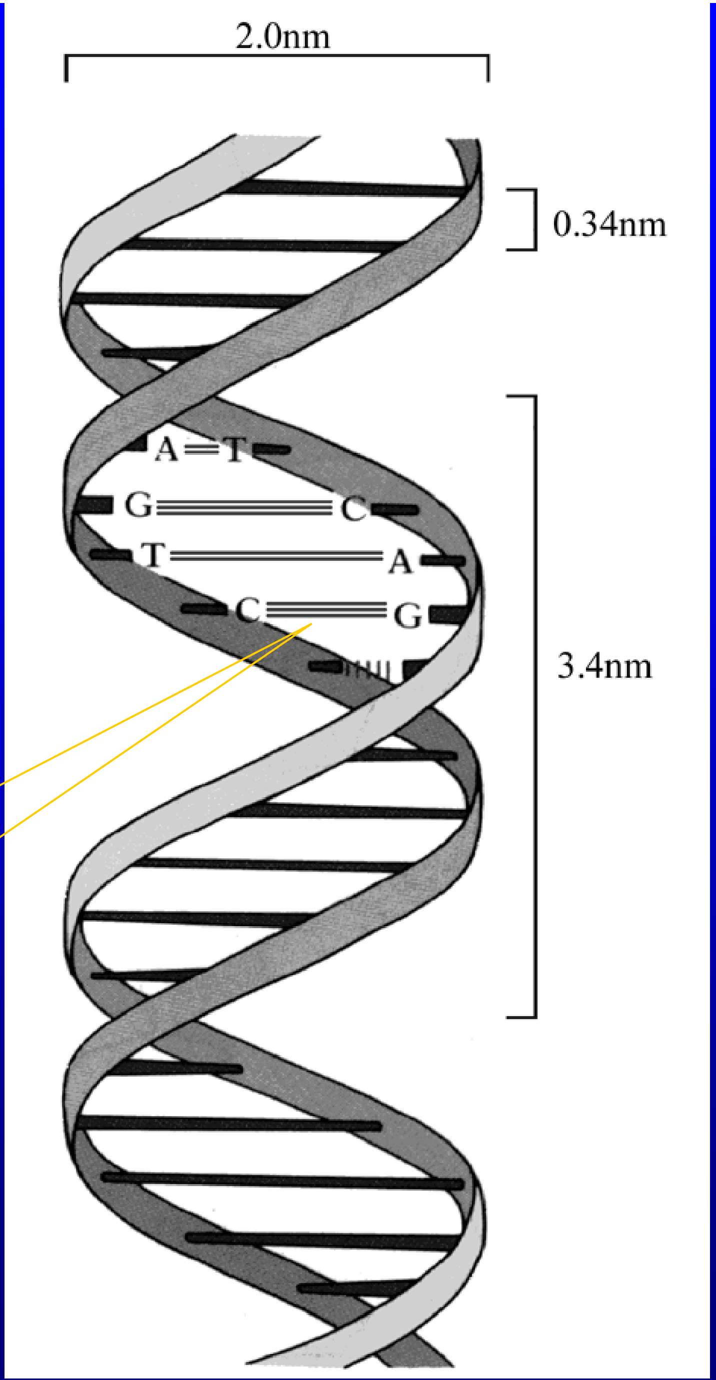
double helix structure

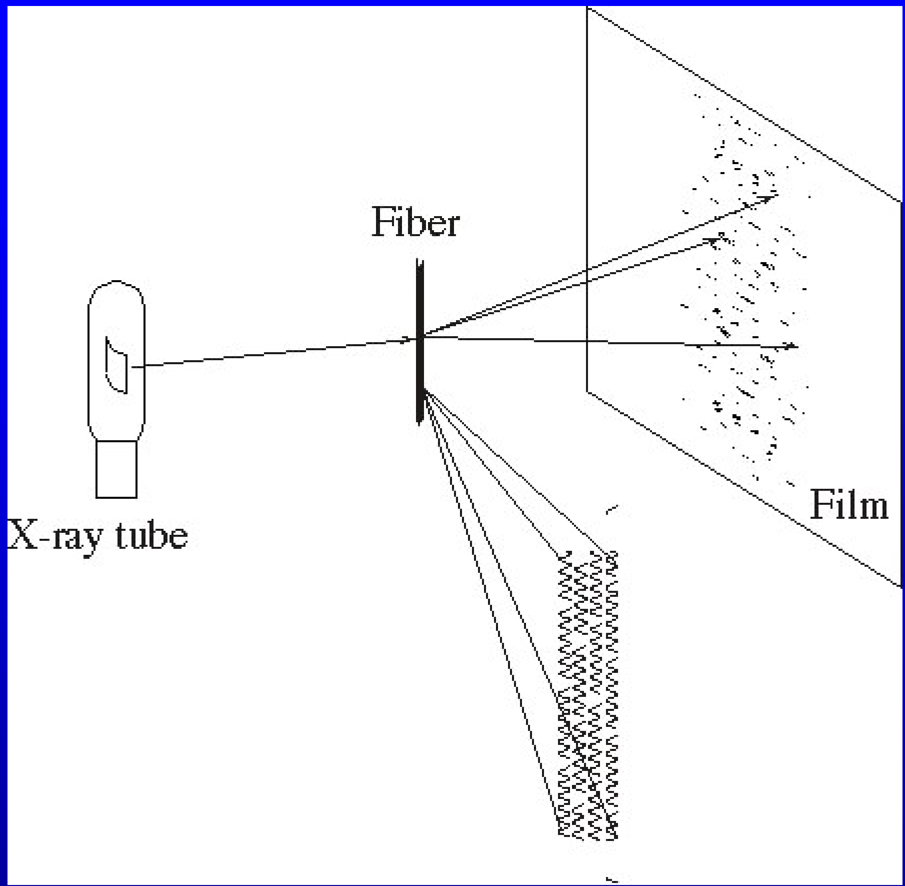
complementary rule:

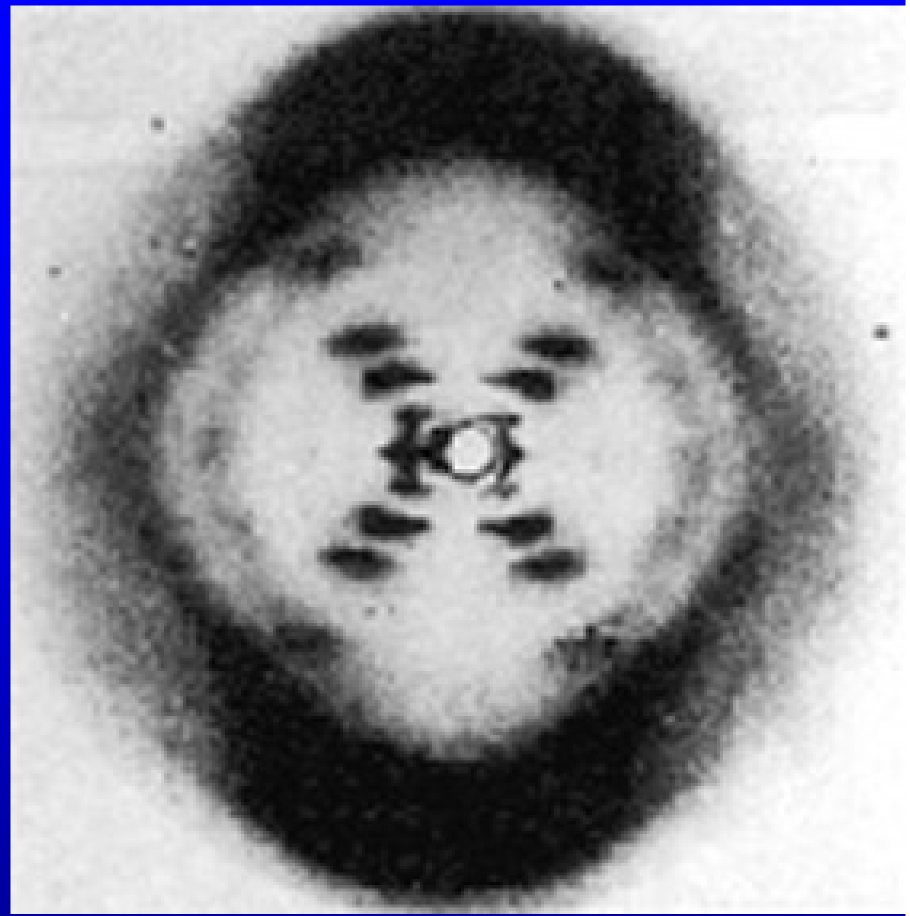
$A = T, G \equiv C, T = A, C \equiv G$

the template theory

Hydrogen bond

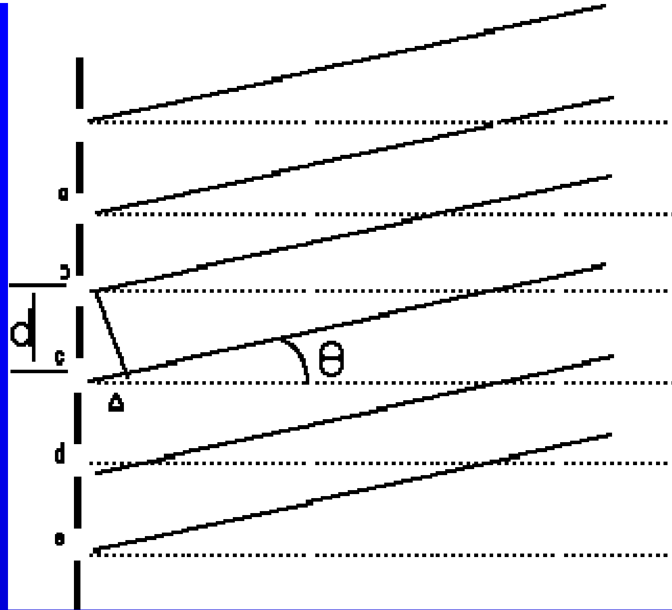


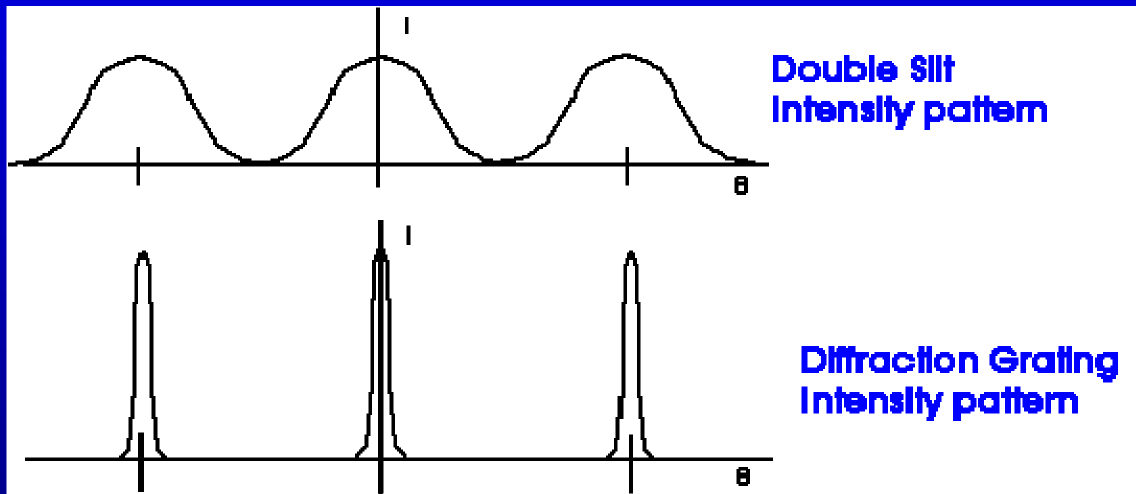




- X-shaped cross
- The fourth layer line of diffraction is missing

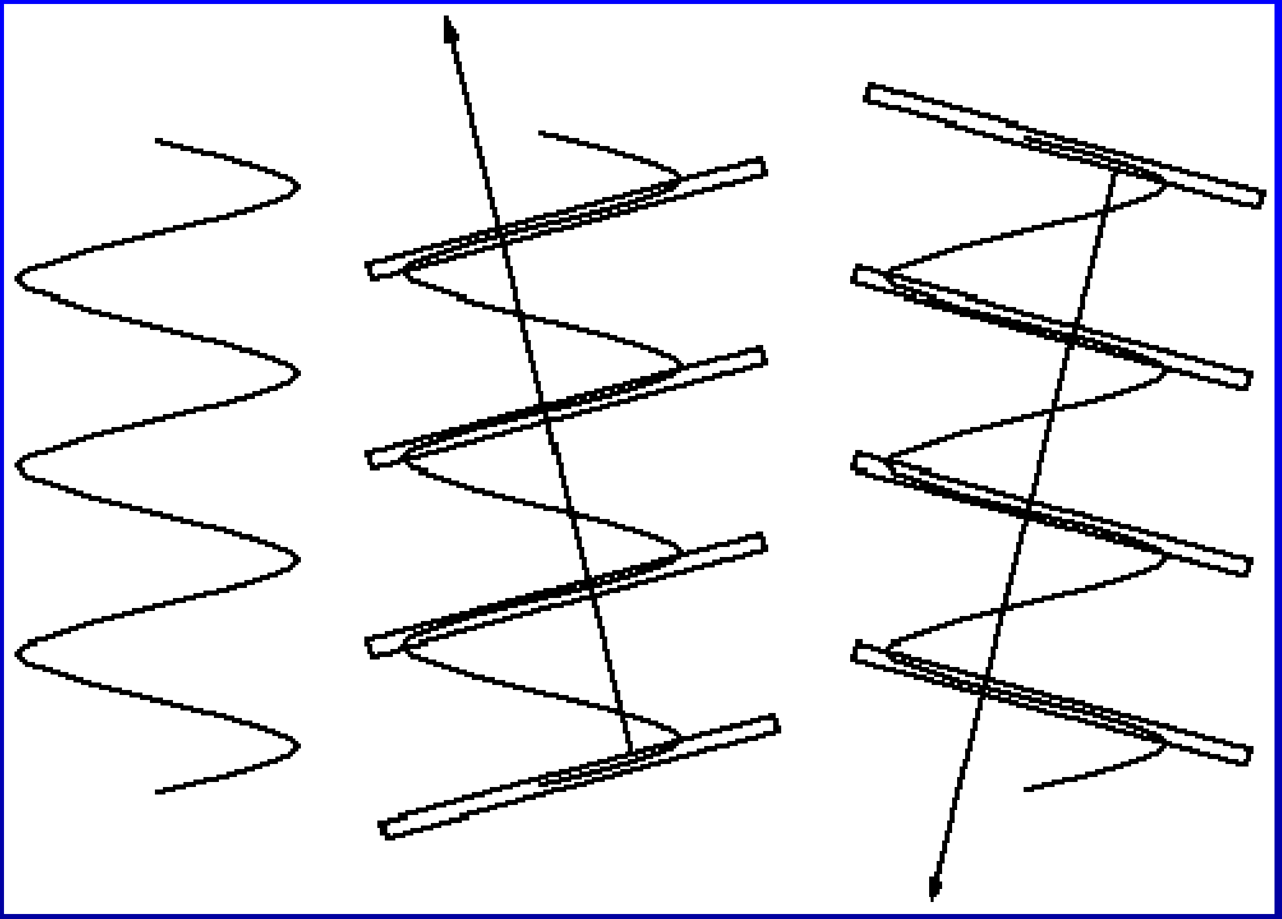
diffraction grating



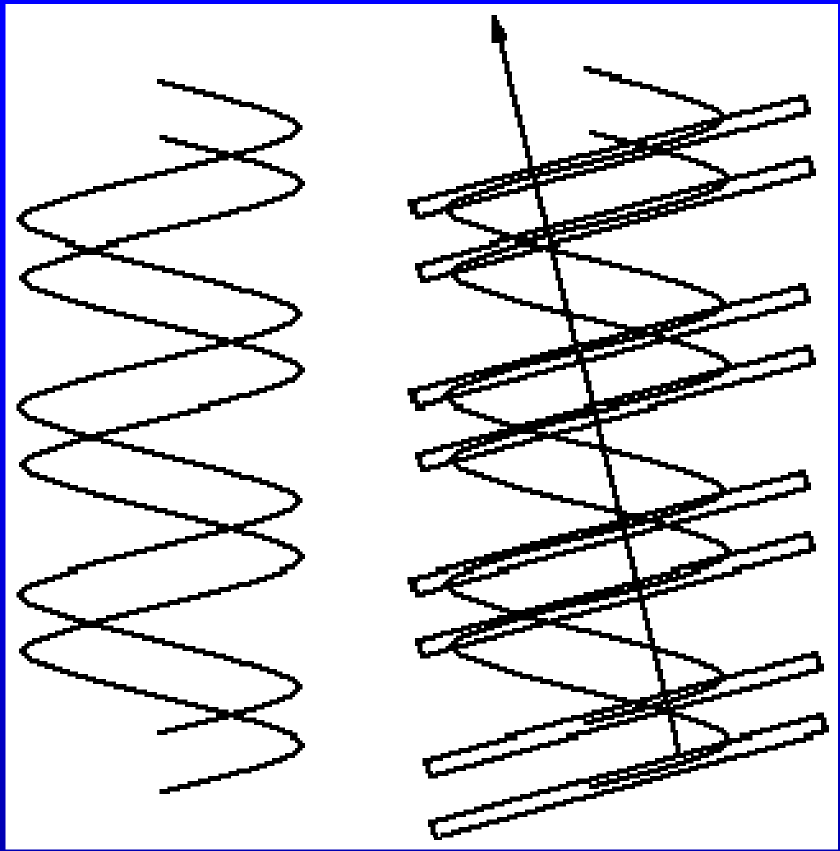


constructive interference for $d \sin\theta = m\lambda,$

$$\mathrm{m} = 0, \pm 1, \pm 2, \pm 3 \cdots$$



•X-shaped Cross



double slit interference

$$\sin\theta = m'\frac{\lambda}{d'}$$ constructive

$$\sin\theta = \left(m' + \frac{1}{2}\right)\frac{\lambda}{d'}$$ destructive

$$d' = \frac{3}{8} d,$$

$$m' = 1$$

$$m' = -2$$

$$\sin\theta = \pm\frac{3}{2}\frac{\lambda}{d'} = \pm 4 \ \frac{d}{\lambda}$$

destructive

Just the position of the fourth layer line

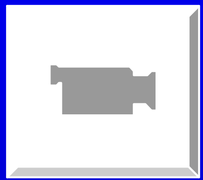
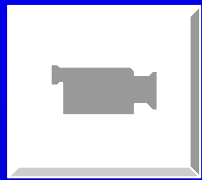
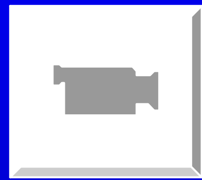
$$\sin\theta = m\frac{\lambda}{d},$$

$$m = 0, \pm 1, \pm 2, \pm 3 \cdots$$

$$m = \pm 4$$

$$\sin\theta = \pm 4 \ \frac{d}{\lambda}$$

Amand Lucas University of Namur (Belgium)

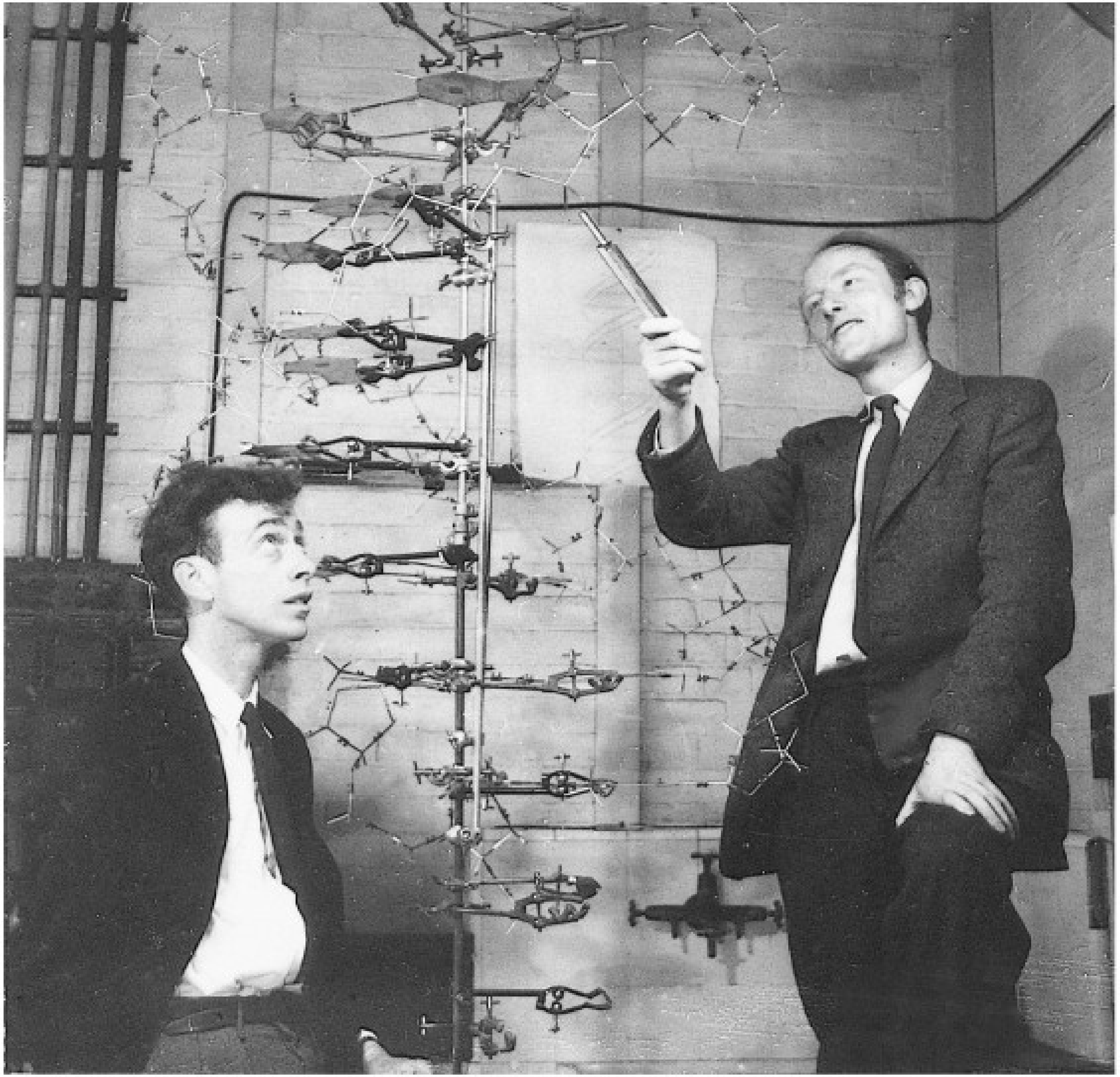
  

Rosalind Franklin

Francis Crick

James D. Watson

Maurice Wilkins



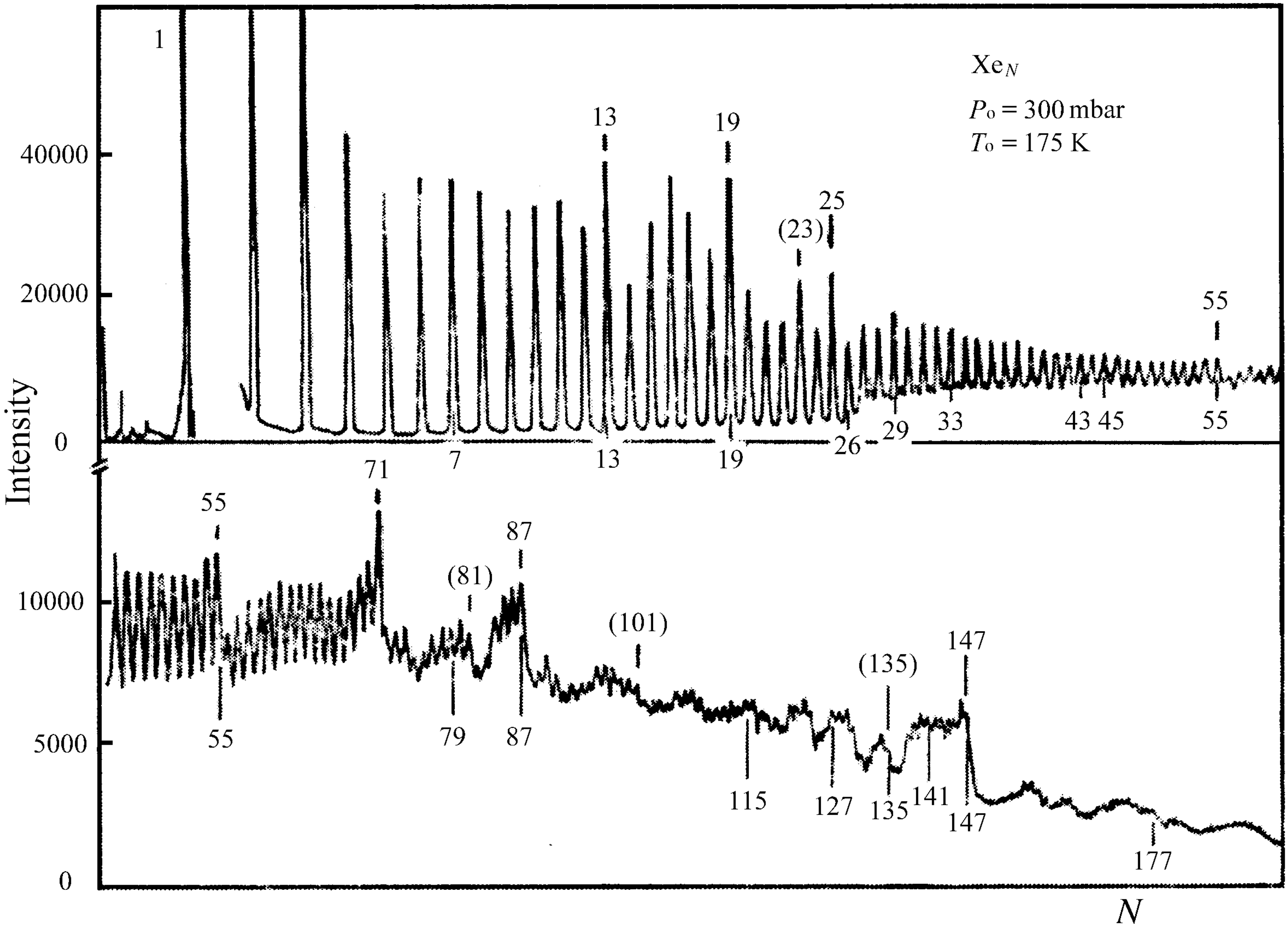
26.6 Clusters

The size of clusters ranges from several to hundreds atoms.

The scale is below 1nm.

Clusters are larger than inorganic molecules， but smaller than bulk solids.

The studies of cluster involve： atomic physics, condensed matter physics, quantum chemistry, surface science, material science, and nuclear physics etc.



Xe$_N$
P_o = 300 mbar
T_o = 175 K
Intensity
40000
20000
0
10000
5000
0
1
13
19
(23)
25
55
7
13
19
26
29
33
43
45
55
71
55
87
(81)
(101)
147
(135)
55
79
87
115
127
135
141
147
177
N

Mass spectrum is used to identify cluster.

The abundance distribution show us thermodynamic stability of clusters

magic number of Xe: 13, 19, 25, 55, 77, 87, 147 etc.

Na:	8, 20, 40, 58, 92
K:	2, 8, 20, 40
Ar:	3, 14, 16, 19, 21, 23, 27
C:	20, 24, 28, 32, 36, 50, 60, 70, 240, 540

Growth spiral

Lennard–Jones
potential

$$U = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$